

Assessing CNN and Semantic Segmentation Models for Coarse Resolution Satellite Image Classification in Subcontinental Scale Land Cover Mapping

Tesfaye Adugna , Wenbo Xu , Jinlong Fan , Haitao Jia, and Xin Luo 

Abstract—Based on studies using high-medium resolution images, convolutional neural networks (CNNs) and semantic segmentation have shown superiority over classical machine learning (ML), particularly in small-scale mapping. However, few/no studies have assessed the techniques on coarse resolution image classification for extensive area land cover mapping. In this study, we evaluated the performance and feasibility of three CNN models (1-D CNN, 2-D CNN, and 3-D CNN), and U-net for coarse-resolution satellite image classification and compared them to a random forest (RF) classifier. We utilized time-series, coarse resolution (1 km) composite imageries acquired by FengYun-3C visible and infrared radiometer. Labeled datasets were collected as shapefiles and split into three independent datasets: training, validation, and test datasets, and preprocessed to meet each model's input format requirements. We conducted several experiments to optimize models and select the best models. Then, the best models were evaluated on an unseen dataset. Among the DL models, one-dimensional (1-D) CNN achieved the highest overall accuracy (OA) 0.87 and kappa (k) 0.84, 2% higher than the best results attained by 2-D CNN, 3-D CNN, and U-net models. However, 1-D CNN is outperformed by RF which achieved 0.89 (OA) and 0.87 (k). Achieving the best and the second-best results using RF and 1-D CNN models, respectively, indicates the superiority of the pixel-based method and the insignificance of spatial information in coarse-resolution image classification. Furthermore, although the DL models can yield high accuracy, especially 1-D CNN, they are less feasible than RF classifiers for coarse-resolution satellite image classification in extensive area land cover mapping.

Index Terms—Convolutional neural networks (CNNs), coarse resolution, deep learning, land cover, semantic segmentation, U-net.

I. INTRODUCTION

THE type of image classification method employed is one of the critical factors that affect the accuracy. Fortunately,

Received 2 March 2024; revised 15 July 2024 and 11 September 2024; accepted 23 September 2024. Date of publication 27 September 2024; date of current version 6 January 2025. This work was supported by National Key R&D Program under Grant 2021YFE0117400. (Corresponding author: Wenbo Xu.)

Tesfaye Adugna is with the Yangtze Delta Region Institute (Huzhou), University of Electronic Science and Technology of China, Huzhou 313001, China (e-mail: atr@csj.uestc.edu.cn).

Wenbo Xu, Haitao Jia, and Xin Luo are with the School of Resources and Environment, University of Electronic Science and Technology of China, Chengdu 611731, China, and also with the Yangtze Delta Region Institute (Huzhou), University of Electronic Science and Technology of China, Huzhou 313001, China (e-mail: xuwenbo@uestc.edu.cn; jhtao@uestc.edu.cn; luoxin@uestc.edu.cn).

Jinlong Fan is with the Geographical Sciences, Beijing Normal University, Beijing 100875, China (e-mail: fanjl@bnu.edu.cn).

Digital Object Identifier 10.1109/JSTARS.2024.3469728

image classification techniques have shown rapid progress and remarkable achievements in recent years, particularly with the advances in machine learning (ML) algorithms and hardware technology. Two decades ago, traditional unsupervised and supervised techniques were the two common classification methods applied to classify remote sensing images with various spatial resolutions to generate local to continental and/or global scale land cover maps (e.g., [1], [2], [3], [4]). In recent years, however, they have been replaced by the most powerful and effective classification methods known as machine learning techniques.

The ML algorithms are nonparametric classifiers [5], [6] that make no assumptions about the distribution of data [7], unlike the traditional supervised classifiers (e.g., maximum likelihood and minimum distance algorithms), which assume the input datasets maintain Gaussian normal distribution that is very unlikely in remote sensing data [6], [8]. Moreover, numerous studies have shown that ML algorithms have achieved more accurate results and demonstrated superior performance in learning complex input datasets with numerous variables/bands and dimensions compared to traditional methods (e.g., [9], [10], [11], [12], [13], [14], [15], [16]). However, these algorithms have some limitations, especially in complex feature/pattern detection compared to deep learning models. They solely rely on pixels' value, i.e., spectral information, for classification [17].

To address the drawback of traditional ML techniques and obtain better accuracy, researchers have suggested and implemented deep-learning techniques, particularly convolutional neural networks (CNNs) and semantic segmentation that consider both contextual and spatial information such as texture, shape, edge, and other attributes of the input data [6], [18], [19], [20], [21], [22], [23]. In addition, the methods require no separate stages/phases of feature extraction and selection procedures, i.e., no manual feature engineering is needed; instead, the models automatically perform feature extraction and classification concurrently [23], [24].

Based on studies carried out using high-resolution imageries, several researchers have demonstrated that DL techniques are more effective and robust than the classical ML algorithms such as support vector (SVM) and random forest (RF) (e.g., [18], [19], [25], [26]). However, most of the studies applying deep learning techniques aimed to classify high-medium (≤ 30 m) resolution remote sensing imageries, including satellite, unmanned aerial vehicle (UAV), and/or aerial imageries,

hyperspectral data, to generate local scale/small area land cover map (e.g., [17], [18], [26], [27], [28], [29], [30], [31], [32], [33], [34]). Nonetheless, the properties/suitability of classifiers are highly related to the attributes of input imageries including its spatial resolution, and extent of the study area which play an important role in controlling interclass and intraclass heterogeneity of the land cover types. Of course, a few researchers have applied the methods for extensive area LC mapping using higher/medium resolution imageries. For instance, Brown et al. [35] generated a real-time global land cover map of eight land cover classes using Sentinel 2 (high-resolution imagery, i.e., 10 m) and fully convolutional neural network techniques. Similarly, Karra et al. [36] employed Sentinel 2 imageries to generate a global land cover map but using a U-net segmentation method that yielded 85% accuracy. However, no investigations have explored the suitability of deep learning techniques in coarse resolution image (>100 m) [37] classification for big area/large scale mapping.

In this study, therefore, we aimed to evaluate the performance of the most commonly applied DL techniques such as one-dimensional (1-D) CNN, 2-CNN, 3-CNN, and U-net semantic segmentation models in coarse resolution (1 km), time-series image classification for subcontinental scale land cover mapping. In addition, we intended to compare the results with RF, one of the best classical ML techniques. To achieve our objective, we processed 9-month, coarse resolution (1 km), 10-day composite imageries that were acquired by FengYun-3C (FY-3C) visible and infrared radiometer (VIRR). Labeled datasets were collected as shapefiles and split into three datasets, i.e., training, validation, and test datasets. Subsequently, the shapefiles were rasterized and preprocessed to meet the input shape format requirements of the different models. Once generating reference datasets, we conducted several experiments using the training and validation samples to find the optimum models. Then, the optimized models were tested on the test dataset for their generalizability, and to evaluate the performance of the models.

The main contributions of this work are as follows.

- 1) We assessed the suitability and effectiveness of four deep learning techniques, 1-D, 2-D, and 3-D CNN, and U-net models for coarse resolution time-series image classifications in extensive area land cover mapping.
- 2) We developed mechanisms to generate reference datasets from shapefiles and input images to train and evaluate DL models.
- 3) This work highlighted how to select the optimum DL models and analyzes the effect of splitting the reference datasets into three independent sets, to train, validate, and test the models.
- 4) This work evaluated the impacts of chip size on the performance of DL models.
- 5) We assessed the efficiency of pixel-based and patch-based classification for coarse-resolution image classification.
- 6) This work compared the performance of 1-D, 2-D, and 3-D CNN, and U-net models, with the classical machine learning algorithm, the RF.

- 7) This study highlights why deep learning techniques (CNN and semantic segmentation) are less effective and suitable for coarse-resolution image classification of big datasets.

II. DEEP LEARNING TECHNIQUES: CNN AND SEMANTIC SEGMENTATION

Deep learning is a learning algorithm consisting of a network of more than one hidden layer of neurons where every neuron is connected across the layer, not within a single layer [38].

Various types of deep learning techniques have been developed and utilized in RS image classification. However, for this work, we assessed the two most commonly applied deep learning models: CNN and semantic segmentation models (U-net). The fundamental difference among these algorithms lies in their architectural designs [39], [40], thus favoring different input dataset formats and classification mechanisms.

A. Convolutional Neural Networks (CNNs)

CNN is one of the most widely used and effective DL models [6], [41], [42], [43]. In the model, a single neuron in the hidden/output layer is connected to a local region/multiple neurons, also known as the receptive field in the inputs layer, unlike a dense neural network (DNN) where every input is connected to every hidden layer. In other words, contrary to DNN, the CNN model is made up of sparsely connected neurons [6], [38]. Besides, in CNN, information is extracted by convolving/sliding windows or matrices of weights over the input image [44]. That is, information about any feature in an image is captured via matrices multiplication between moving weights matrices (filters) and an input image, not through direct multiplication of vectors without convolution as in NN [20], [45]. These two unique attributes of CNN, sparse neuron connections and convolving weight matrix, not only allow for parameter (weight) sharing which reduces the number of trainable parameters in CNN models, but also enable the model to capture spatial information, making it more efficient and accurate than DNN [6], [17], [20], [39].

Generally, CNN architecture commonly consists of three connected main layers [44], [46], [47] (see Fig. 1) that perform different functions (see Fig. 2). These layers are the convolutional layer, pooling layer (subsampling or downsampling layer), and fully connected layer; plus, nonlinearity activation functions that are associated with the convolutional and fully connected layers.

The convolution layer extracts information/feature maps by convolving filters/kernels, which are also known as matrixes of learnable parameters (weights), of neurons connected to the input. In other words, the convolution layer performs element-wise multiplication between the weight matrix (the moving filter) of neurons and subregions in the input layer [23].

Then, the activation nonlinearity functions, commonly ReLu function [44], convert feature maps obtained by convolution/multiplication operations to non-linear functions so that the model learns complex/nonlinear relations [48].

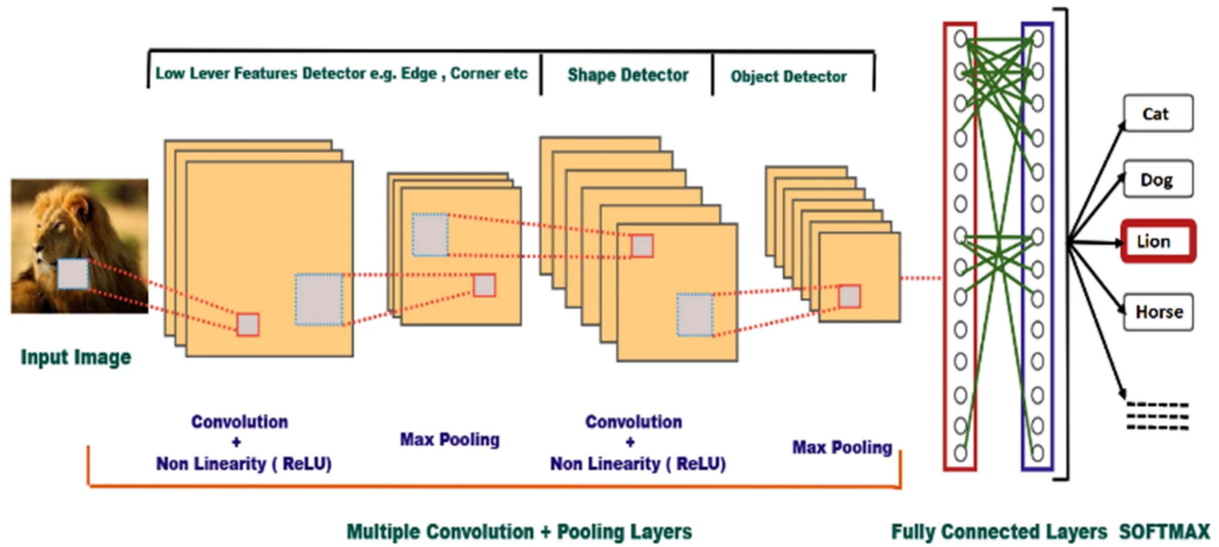


Fig. 1. Typical CNN architecture. From [44].

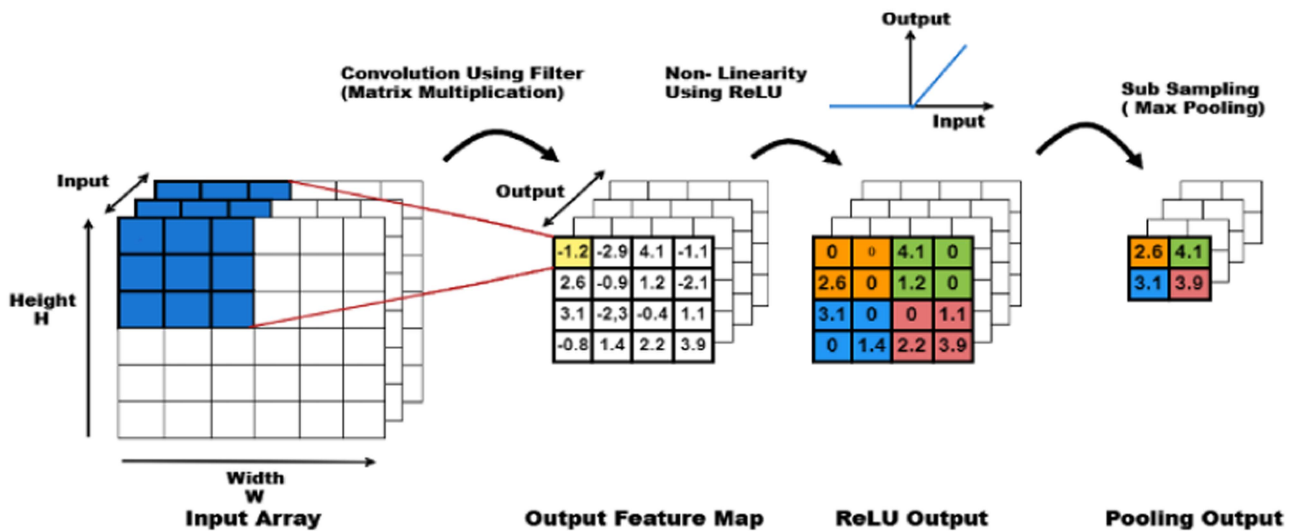


Fig. 2. Role of convolution, max-pooling, and ReLU functions during feature map generation in CNN. From [44].

The pooling layers are used to reduce the dimension (height and width) of the feature map by taking the max/average of pixel values under a striding widow, which has a defined pooling size, on the feature map [24], [46]. The function of pooling layers is to summarize the feature map by finding the maximum or mean values of feature maps' pixels, thereby minimizing computational time and overfitting at the expense of spatial information [48].

At the end of the convolutional layers, the model is flattened, i.e., the 2-D feature maps are converted into 1-D (vector) and transferred to the fully connected layers where every neuron is connected to every activation in the previous layer (see Fig. 1) in which case no parameter sharing takes place like the regular neural network. The activation for every neuron in the fully connected layer is calculated using the same technique as NN, i.e., $a = \sigma(w^T x + b)$, and followed by a nonlinearity function

that applies an element-wise operation to convert the activation to non-linear form [24].

At the end of the network, the Softmax activation function is used to normalize activation scores, i.e., to make their sum equal to one, and compute their probability distribution to the respective classes; in other words, the Softmax predicts the probability of every class using the activation scores from the final connections [49], [50], [51].

CNN can be categorized as 1-D CNN, 2-D CNN, and 3-D CNN based on the type of convolutional layer employed [30], [42]. In 1-D CNN architecture, the model employs 1-D convolutional and pooling layers, while, in 2-D CNN and 3-D CNN the convolutional and pooling layers become 2-D and 3-D, respectively. As a result, the models require different input shapes, i.e., 1-D CNN expects 3-D tensors, (batch size, timestep, channels); while 2-D CNN works on 4-D tensors (batch size,

height, width, channels), and 3-D receives 5-D tensors (batch size, height, width, depth, channels).

CNN can also be classified as a pixel-based classifier (1-D CNN) or patch-based (2-D CNN and 3-D CNN) on the basis of the final output type they yield. The pixel-based classifier makes pixel-wise classification; whereas, the patch-based models predict on a chip level by assigning individual chips to a single class based on the highest probability score.

Furthermore, CNN models can be considered a shallow or deep model-based number of layers/depth in the main layers [23], [50]. However, the selection of the number of layers/depth of the model is subjected to parameter tuning along with other CNN hyperparameters including, the number and size of kernels, and regularizes to obtain the best results [20], [23], [24], [38].

B. Semantic Segmentations

Semantic segmentation is a pixel-based classification where input datasets are prepared and fed as patches with a certain size, and the model classifies each pixel in the patch to its respective classes based on their common semantic characteristics [20], [21], [52]. Semantic segmentation models map each pixel to its corresponding class at the last stage, like classification.

However, semantic segmentation and classification are essentially different techniques. In semantic segmentation, the model segments/divides the input image/patch into distinct regions based on their features extracted by the sliding window; subsequently, the pixels in the segmented region are assigned to a class with the maximum probability. In the latter case; however, pixels are directly categorized into one of several classes on the basis of the pixels' values without undergoing segmentation, [53]. Applying semantic segmentation methods offers several advantages over patch-based CNN models. In semantic segmentation, each pixel in a chip is predicted individually, unlike in CNN where each pixel is assigned the class of its enclosing patch or region [54]. This means that every pixel in the patch is labeled as the same class, even if some pixels in the patch represent different classes, which could lead to misclassification. Additionally, semantic models not only yield better accuracy than CNN models (e.g., [18], [29], [52], [55], [56]) but also allow us to overcome the limitation of CNN models related to the loss of spatial resolution/information [40], [57].

Various types of semantic segmentation models have been developed and applied in RS image classification (e.g., DeepLab V3+ [17], DeepLab [57], fully convolutional network [58], U-net [59], and SegNet [60]). These models have some common features in terms of overall architecture; they all have an encoder (convolutional block) and a decoder (deconvolutional block, yet they modify or use different CNN backbone models that result in the variation of their encoder block (depth) [39], [61] and most profoundly, they differ in decoding, training and inference [60].

Although the U-net is a relatively older model, it is widely employed due to its high level of performance [18], [23], [28], [52]. Therefore, we chose the model for our work.

1) *U-net*: A U-net is a segmentation technique that comprises two sections/paths: the convolution/contracting section to extract the contextual information, and the expanding/

deconvolution path that allows accurate localization [39]. In the deconvolutional section, in addition to up-sampling, the model combines layers to integrate low-level and high-level information/features, aiding in recovering lost information, including spatial resolution and semantic details [28], [59]. In other words, high-to-low-resolution subnetworks are connected in series and restore the resolution via an up-sampling process, i.e., low-high up-sampling [62].

III. MATERIAL AND METHODS

A. Input Data

We received a one-year (April 2019 to March 2020), coarse resolution (1 km), FY-3C VIRR composite imageries from the National Satellite Meteorological Center of China. Out of the one-year data, we systematically selected 9-month imageries based on our previous findings [63]. The imageries comprise 297 band metrics composed of 11 distinct bands, which include 4 visible, 3 IR, 3 TIR, and the NDVI. To manipulate all 297 band metrics/channels as a single image, the images were stacked in ENVI 5.3 software.

B. Labeled Data

We collected labeled datasets as regions of interest (ROIs)/polygons representing eight land cover classes (bare-sparse vegetation, built-up, cropland, forest, herbaceous wetland, herbaceous vegetation, shrub, and water body), across the study area (see Fig. 3). The details of the labeling/class naming and collection mechanisms were discussed in our previous work [63]. However, for the sake of completeness, a brief description is given as follows. The ROIs (polygons representing a particular class) were annotated in ENVI software by employing three techniques parallelly, i.e., Landsat 8 image interpretation, cross-checking and consulting an existing map, and Google Earth. Once the annotation was completed, the ROIs were saved as a shapefile. Then, the shapefile was systematically divided into three datasets for training, validation, and testing, with a ratio of 68%, 20%, and 12%, respectively, where each dataset comprised all of the eight land cover classes. However, since the shapefiles cannot be used directly as reference data, they were further manipulated to meet each model's reference dataset format/ shape requirements.

C. Hardware and Software

For this work, a laptop computer with a 2.7 GHz processor 32 GB RAM, and NVIDIA GeForce RTX 3050 Ti Laptop GPU. The GPU has 4 GB dedicated memory and 16 GB shared memory were employed. We used Python for data preprocessing, model training and evaluation, classification, and postprocessing. Python libraries/modules such as osgeo, gdal, ogr, numpy, sklearn, TensorFlow, keras, pickle, matplotlib, geopandas, and others, were employed to perform the various tasks. In addition, ENVI 5.3 and Arcmap 10.3 were utilized for pre and postprocessing operations, including training data collection and ROI/shapefile manipulation, image stacking, and output visualization.

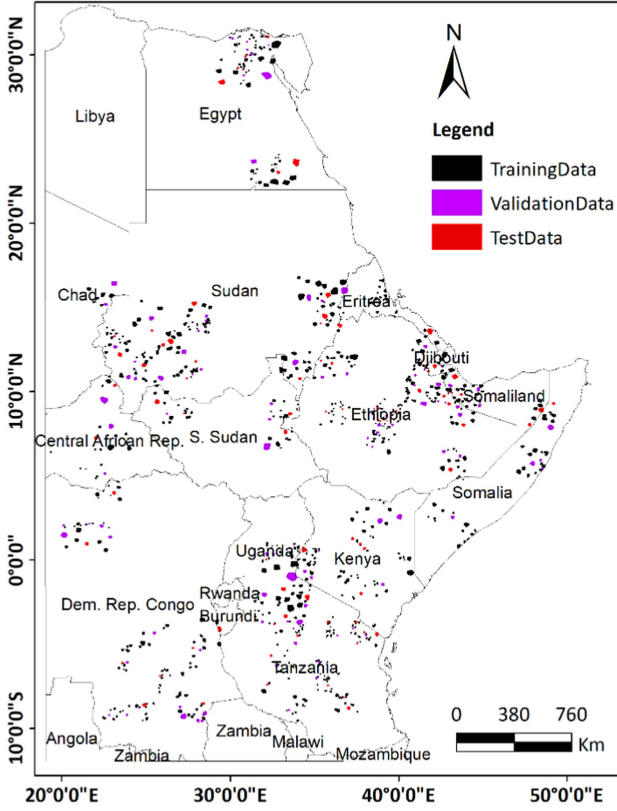


Fig. 3. Labeled data distribution across the study area.

D. Technical Workflow

We implemented two distinct technical routes for CNN and semantic segmentation models (U-net) (see Figs. 4 and 5) as the two models expect different reference dataset formats. The difference between the two workflows lies in how the reference datasets were generated for the two models, except that the overall approach is similar. To highlight the main distinctions in the workflows, four boxes in the U-net model workflow, Fig. 5, were colored blue.

E. Preprocessing: Reference Datasets Generation

As briefly stated in Section III-B, different models require different input and reference data shapes. Thus, we implemented three strategies to generate reference datasets for pixel-based models (1-D CNN and RF), batch-based models (2-D CNN and 3-D CNN), and U-net models.

1) *Reference Dataset Preparation for Pixel-Based Models:* To prepare the reference datasets for pixel-based models (1-D CNN and RF), the three shapefiles, generated in Section III-B, were rasterized to create three mask layers comprising labeled pixels if they are encircled by the ROI/shapefile else zero, i.e., unlabeled or masked out region. Subsequently, the unmarked pixels were eliminated by applying basic mathematical logic to the mask layers (i.e., $y = y [y > 0]$), removing all pixels with zero values and retaining the remaining pixels with values ranging from 1 to 8, representing the class labels. Finally, the filtered mask layer (a 1-D tensor of labels) is utilized to

slice the corresponding input image (X). This process yields output datasets (X, y), where X represents the input image in the 2-D tensor form, and y denotes the corresponding labels in 1-D tensor/array format that is suitable for direct utilization in the RF model (see Table I). However, in 1-D CNN they should be reshaped to meet 1-D CNN model input shape requirements, i.e., X is converted into 3-D tensors by expanding its dimension, and y is converted to 2-D tensors by converting them into categorical (see Table I).

2) *Reference Data Generation for Patch-Based Models:* In patch-based CNN models (2-D CNN and 3-D CNN), the association between reference images/patches and their labels is one-to-one, i.e., each reference image corresponds to a single land cover type/class. As a result, with three distinct reference datasets (training, validation, and test), each comprising 8 classes, we must create a reference image for every class, totaling 24 reference images. Accordingly, we initially split the three shapefile datasets created in the previous Section III-B, each with 8 classes, into 24 distinct shapefiles each with a single class. Subsequently, all shapefiles were converted into raster format to produce 24 mask layers consisting of pixels with values of 1 and 0 to denote labeled areas and unlabeled regions, respectively. These mask layers were then utilized to mask the primary input image (SIIMG). As a result, the spectral values and spatial information of SIIMG pixels corresponding to mask layer pixels with a value of 1 were retained, while the remaining pixels were zeroed out and masked. This approach was employed to generate a total of 24 masked reference images for the three separate reference datasets (training, validation, and testing), with each dataset containing 8 distinct masked reference images representing the 8 classes.

Finally, each masked reference image was chipped into tiles/patches, sorted out to separate patches with image information from completely masked-out regions/pixels, and relabeled back to the class it represents, as described in the following subsection.

a) *Chipping, sorting, and relabeling patches:* Chipping masked reference imageries (MRImg), are one of the vital preprocessing stages for two main reasons. First, chipping the masked reference imageries (MRImg) enables us to sort the labeled regions (unmasked) and unlabeled regions/pixels in the MRImg.

Second, when dealing with multiclass classification tasks for a single input image using patch-based models, it is essential to divide the masked reference images and the main input image (SIIMG) into appropriate tile/chp size. Particularly, when working with coarse-resolution input images, careful consideration should be given to the patch size because dividing the reference datasets and input images into large patches increases pixel heterogeneity within the patch, as it contains various land cover types into a single chip. This can result in minor classes, represented by only a few pixels and covering small areas, being overlooked or not being identified as separate independent classes. Instead, they will be homogenized into the majority class and classified as part of the dominant class within the patch, as patch-based models tend to assign a chip to a single class. In other words, chipping coarse resolution input images

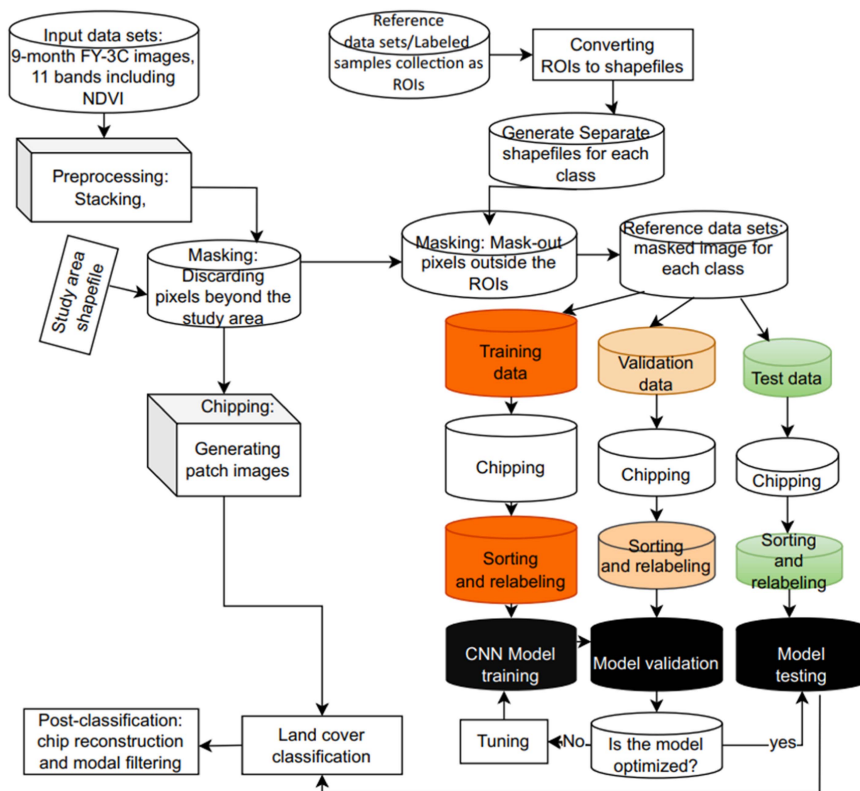


Fig. 4. Technical routes for patch-based CNN model.

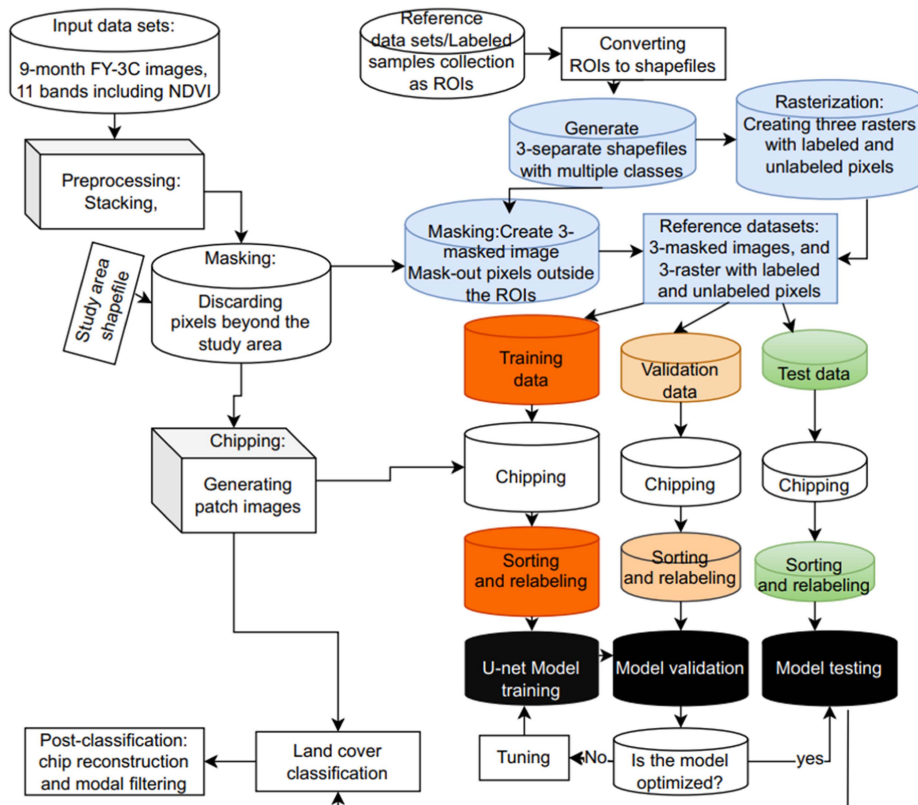


Fig. 5. Technical routes for segmentation model.

TABLE I
REFERENCE DATASETS FOR PIXEL-BASED CNN AND ML MODELS

Model	Training data	Validation data	Test data
1D CNN	X_train: (66100, 297, 1)	X_val: (20057, 297, 1)	X_test: (11751, 297, 1)
	y_train: (66100, 8)	y_val: (20057, 8)	y_test: (11751, 8)
RF	X_train: (66100, 297)	X_val: (20057, 297)	X_test: (11751, 297)
	y_train: (66100,)	y_val: (20057,)	y_test: (11751,)

TABLE II
REFERENCE DATASETS FOR PATCH-BASED CNN MODELS

Model	chip size	Training data	Validation data	Test data
2D CNN	2 x 2	X_train: (20412, 2, 2, 297)	X_val: (5746, 2, 2, 297)	X_test: (3451, 2, 2, 297)
		y_train: (20412, 8)	y_val: (5746, 8)	y_test: (3451, 8)
	3 x 3	X_train: (10964, 3, 3, 297)	X_val: (2910, 3, 3, 297)	X_test: (1779, 3, 3, 297)
		y_train: (10964, 8)	y_val: (2910, 8)	y_test: (1779, 8)
	5 x 5	X_train: (5415, 5, 5, 297)	X_val: (1325, 5, 5, 297)	X_test: (838, 5, 5, 297)
		y_train: (5415, 8)	y_val: (1325, 8)	y_test: (838, 8)
3D CNN	2 x 2	X_train: (20412, 2, 2, 27, 11)	X_val: (5746, 2, 2, 27, 11)	X_test: (3451, 2, 2, 27, 11)
		y_train: (20412, 8)	y_val: (5746, 8)	y_test: (3451, 8)

into large-size patches can lead to misclassification/error during prediction, particularly the minor classes. For example, in a heterogeneous region, a 5×5 chip (covering 25 km²) is highly likely to contain two or more different classes with different proportions. In such circumstances, the models homogenize the pixels and predict/categorize them to the dominant class in the chip, which leads to misclassifications.

Therefore, we prepared three sets of patches with sizes of 2×2 , 3×3 , and 5×5 (see Table II) to select the best tile size and analyze the impact of patch size on the performance and the output map or classified image. Once the patches were generated, we sorted out the patches with information (spectral/spatial) from those without useful information (masked-out chips).

Finally, every chip containing valuable information was relabeled to match the original land cover types (classes), which is the name of the ROI/shapefile. During relabeling, to give a unique record/name for each chip, we used 6-digit floating numbers (e.g., 1.090000) where the number before the decimal (1) indicates the class it represents; while the number after the decimal (.090000) denotes the uniqueness of the chip. The decimal numbers are set to update by minus 1(-1) so that chips are arranged consecutively. For example, for cropland (class 3), sorted chips were relabeled as cropland.3.090000, cropland.3.089999.npy, cropland.3.089998.npy, and so on. Similarly, for built-up (class 2) labels were generated as, built-up.2.090000, built-up.2.089999.npy, built-up.2.089998.npy, ..etc. Finally, when the chips/reference images are fed to the models, the numbers before decimals are extracted and used as the labels (y), and the chip is read as the input image (X).

3) *Reference Data Generation for Segmentation Model, U-Net*: Segmentation models expect the labels as separate rasters, unlike patch-based CNN models where the labels are

attached to the chips. In other words, segmentation models require two separate reference rasters to train the models: the images/chips and their corresponding labeled mask.

To generate the labeled masks, we rasterized the three sets of shapefiles prepared above, in Section III-B. Consequently, we created labeled masks containing pixels with class labels (1 to 8) if they are within the ROI/shapefile and zero in unlabeled/masked-out regions. Subsequently, the labeled masks were divided into tiles and sorted out to distinguish patches with labels from those without.

To prepare the input images/patches, we masked the main input image (SIIMG) using three mask layers consisting of pixels with values of 1 and 0 representing labels and unlabeled regions, respectively. As a result, three masked input images were generated. In the masked input images, pixels that align with mask layer pixels with a value of 1 retain the spectral values and spatial information, while the remaining pixels are set to zero (0), effectively masked out. Subsequently, the masked input images were patched to the same size as the corresponding labeled mask, and sorted out to separate patches with image information from completely masked-out regions/pixels

F. 1-D, 2-D, and 3-D CNN Model Training and Optimization

In deep learning model training, performing parameter tuning and regularization is crucial to prevent/minimize overfitting, enhance their generalizability, and achieve the best result [24], [38], [64], [65]. Therefore, to optimize the three CNN models and obtain the best architecture of each model, we conducted several experiments using the training and validation datasets (see Table II) and employed common parameter tuning and regularization methods.

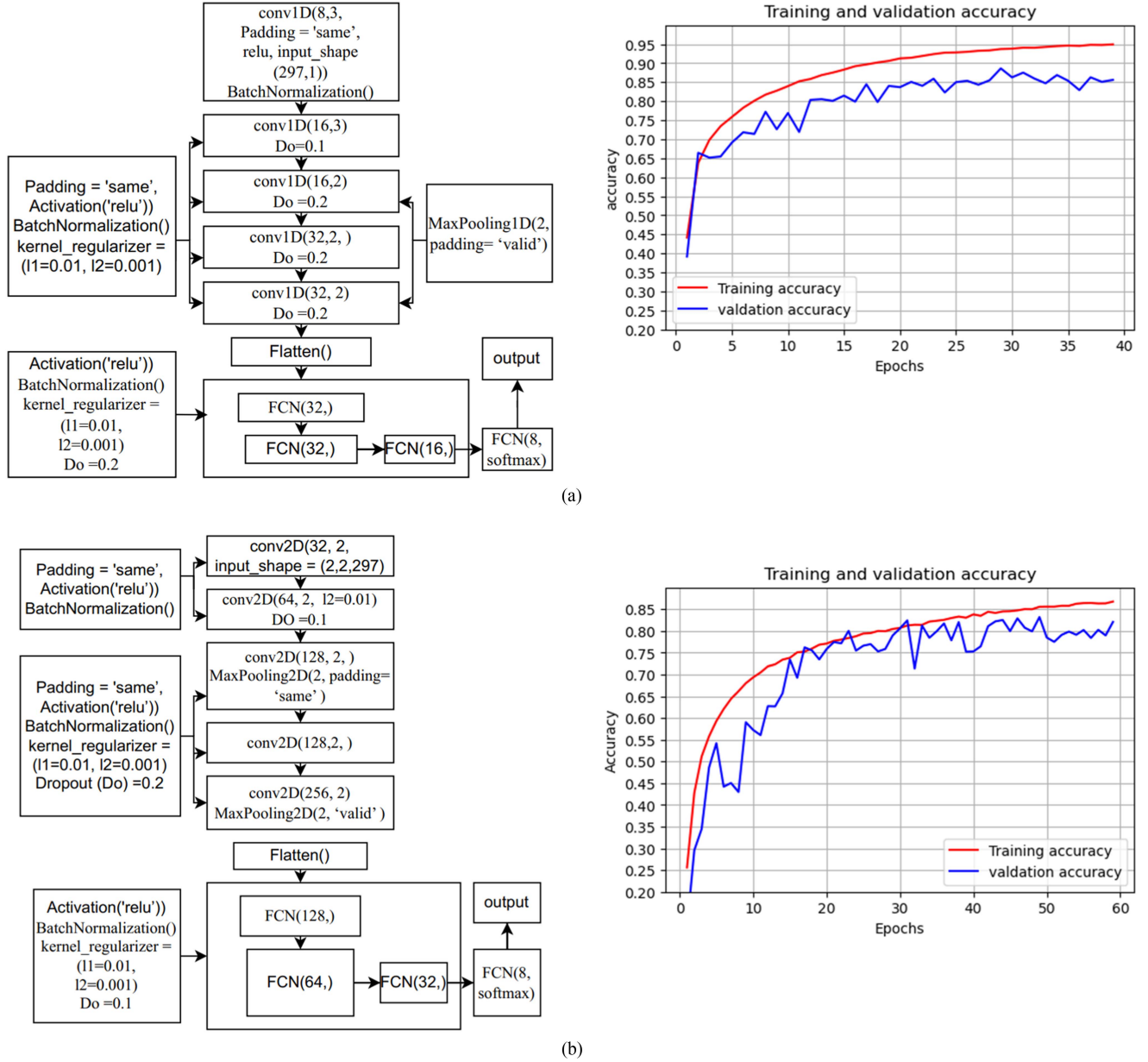


Fig. 6. Best architectures of the various CNN models (1st column) and their corresponding performance plots (2nd column). (a) 1-D CNN. (b) 2-D CNN using 2×2 input patch.

These methods include adjusting the model's depth, selecting the kernel size and amount, applying padding and batch normalization, as well as determining the size and position of pooling, dropout, and l1 and l2 regularizes.

Additionally, we have explored different batch sizes, utilized early stopping, and adjusted the learning rate for model optimizers, specifically the Adam optimizer, which is widely used in deep learning [24]. Finally, the optimal architecture for each model (see Fig. 6) was chosen based on overall accuracy on the validation dataset and train-validation accuracy plots (see Fig. 6).

Once the models were optimized, they were evaluated on test datasets (see Table II) to assess their ability to generalize on unseen data, and used for classification/prediction purposes. The

models were quantitatively evaluated using accuracy measuring metrics such as confusion/error matrix, recall, precision, and OA, the most commonly employed techniques to quantify the accuracies of various classifiers including deep-learning models [42], [66]

G. U-Net Model Training and Optimization

Two U-net models were trained using the training datasets chipped into different sizes, 3×3 and 5×5 (see Table III), so as to analyze the impact of patch size and compare the results with CNN models. To obtain the optimal models' architecture (see Figs. 7 and 8), we finetuned the parameters and hyperparameters following similar strategies employed for optimizing

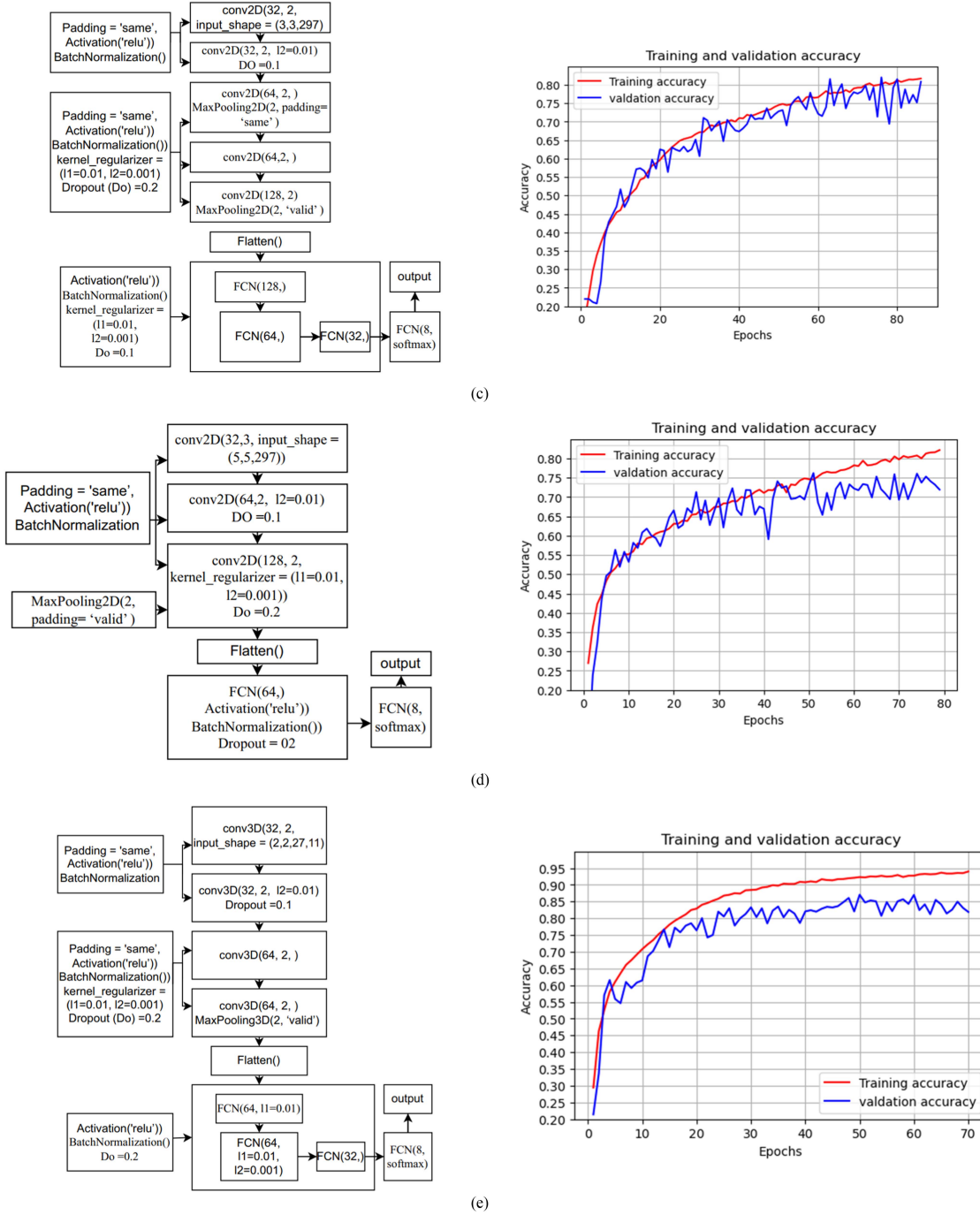


Fig. 6. (Continued.) Best architectures of the various CNN models (1st column) and their corresponding performance plots (2nd column). (c) 2-D CNN using 3 × 3 patch. (d) 2-D CNN using 5 × 5 patch. (e) 3-D CNN using 2 × 2 input size.

TABLE III
REFERENCE DATASETS FOR U-NET MODELS

Model	chip size	Training data	Validation data	Test data
U-net	3 x 3	X_train: (7344, 3, 3, 297)	X_val: (2228, 3, 3, 297)	X_test : (1305, 3, 3, 297)
		y_train: (7344, 3, 3, 8)	y_val: (2228, 3, 3, 8)	y_test: (1305, 3, 3, 8)
	5 x 5	X_train : (2644, 5, 5, 297)	X_val : (802, 5, 5, 297)	X_test: (470, 5, 5, 297)
		y_train: (2644, 5, 5, 8)	y_val: (802, 5, 5, 8)	y_test: (470, 5, 5, 8)

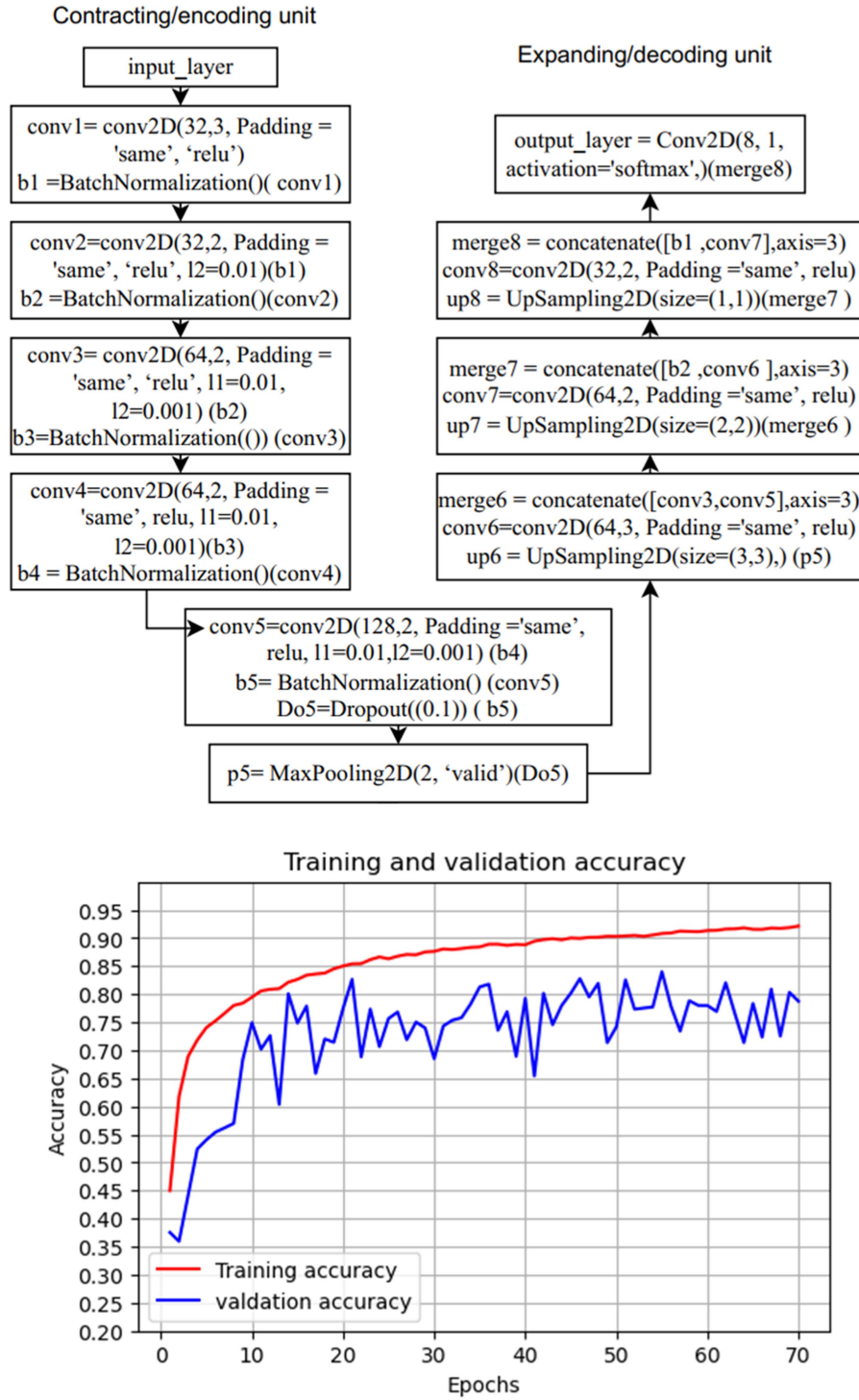


Fig. 7. Optimum architecture for U-net using 3×3 input chips (upper) and its performance plots (lower).

CNN models. Then, the models were evaluated on test datasets and utilized to construct the map, i.e., predict the unknown.

H. Random Forest (RF)

RF is an ensemble classifier that employs a majority vote after constructing several decision trees using randomly selected

predictors/features [67], [68]. To compare the performance of the advanced DL models (CNN and U-net) described above with the pixel-based classical ML model, we employed RF. The classifier was chosen not only because it is one of the most robust ML classifiers [67], but also because it outperformed SVM in our previous work using the same datasets [69]. We used default parameters provided in Sklearn library [70] based

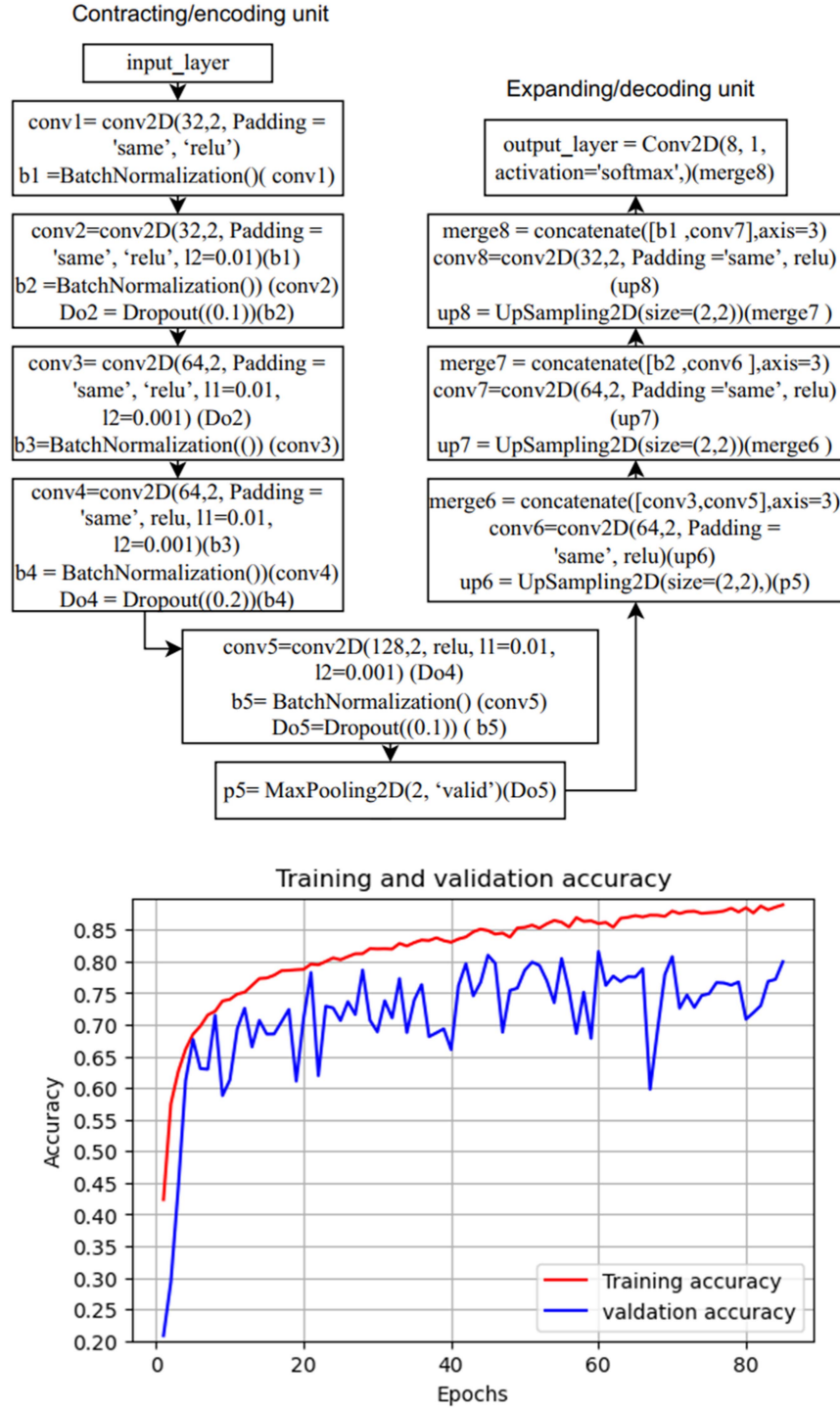


Fig. 8. Optimum architecture for U-net using 5×5 input chips (upper) and its performance plots (lower).

on our previous work [69] to train the model. However, for this particular case, we utilized three reference datasets (train, validation, and test) which are the same datasets used for 1-D CNN (see Table I)

I. Postprocessing

After classifying the unknown input pixels/chips in the study area, we sequentially conducted two crucial postprocessing

steps. This involved reconstructing the predicted chips/pixels to generate the full land cover map of the study area followed by modal filtering to eliminate noise. In pixel-based classification (RF or 1-D CNN), reconstructing an output/classified pixel to its original input image is straightforward; it is simply done by reshaping the predicted vectors into a 2-D shape.

However, constructing predicted chips/patches back to the original input image shape and size is tricky and challenging.

Moreover, patch-based CNN (2-D CNN and 3-D CNN) and segmentation (U-net) models operate differently and yield different types of outputs, i.e., CNN assigns one class for a chip, whereas segmentation classifies the patch pixel-wise. Consequently, we employed two different approaches to reconstruct the predicted chips from the two models.

To reconstruct predicted chips from the CNN models, first, we initialized an out-put image (zeros) with the same size as the original image, then iterated over the predicted chips to determine row and column indices of each block, and filled in the output image with the blocks and predicted classes.

For the U-models, to reconstruct predicted chips, first, we reshaped the predicted chips in (row, column) format, created an empty output array with the same shape as the original image, then looped over the chips and filled in the output array with original labels that were obtained by mapping the predicted labeled to the original labels. After image reconstruction, a modal filter with a 3×3 moving window was applied to remove/reduce the salt and pepper effect from the final Lc product.

IV. RESULTS

As shown in Tables IV, V, and Fig. 9, the results revealed that the best model of each DL classifier (CNN and U-net) achieved high accuracy ($> = 85\%$) despite significant performance variations due to the type of the models and/or size of chips input. Among the DL models, 1-D CNN achieved the highest accuracy both on validation (OA = 0.89, $k = 0.86$) and test datasets (OA = 0.87, $k = 0.84$), which are 5%–4% and 3%–2% higher than the best results attained by 2-D CNN and U-net models on the same reference datasets.

The best 2-D CNN model yielded an OA = 0.84 and $k = 0.81$ on validation samples, and an OA = 0.85 and $k = 0.82$ on test datasets when the smallest chip size (2×2) was manipulated. On the same validation datasets, 3-D CNN achieved a 1% higher accuracy (OA and k) than 2-D CNN but the same OA and k on test samples. On the other hand, the best U-net model achieved similar accuracy as 2-D CNN when 3×3 chips were manipulated, implying the patch-based CNN models and segmentation model can yield comparable overall accuracy if the right size, i.e., smaller chips datasets are processed during coarse resolution image classification.

However, processing larger patches has negatively impacted the accuracy in both patch-based CNN and U-net models. In 2-D CNN, when the size of chips increased from 2 to 5 the accuracy dropped by 9% and the model yielded the least OA (0.76) and k (0.71). In U-net, increasing the patch size from 3 to 5 lowered the accuracy by 3% on the validation datasets, and by 1% on test data, which indicates U-net models are less impacted by increasing the patch size compared to patch-based CNN models.

Moreover, the results demonstrated that the RF model achieved the best accuracy (OA = 0.89) and ($k = 0.87$) on test datasets by outperforming 1-D CNN which attained the highest accuracy among the DL models. However, on validation datasets 1-D CNN achieved a 3% higher OA and k than RF.

V. DISCUSSION

A. Comparison and Evaluation of Models

As indicated in Tables IV and V, among the DL techniques, a 1-D CNN model outperformed all the patch-based models (2-D, 3-D,) and U-net segmentation techniques. Nonetheless, the model was outperformed by the RF classifier. In other words, among all the models considered in this study, the RF model achieved the highest accuracy (OA and k) on test datasets, but lower accuracy than 1-D CNN on validation datasets. This could be due to the leakage of some validation samples during repetitive parameter and/or hyperparameter tuning operations that were not performed for the RF. During RF model training, no parameter tuning was performed, instead, we employed default parameters as given in the sklearn library [70] based on our earlier finding [69].

Therefore, based on their performance on test datasets, RF is considered the best model implying the superiority of classical ML methods over the advanced ML techniques for coarse-resolution image classification as opposed to the previous reports based on high-resolution image classification [18], [19], [25], [26], [47], but in agreement with the results obtained by [20] using medium-resolution imageries.

With regard to 1-D CNN, our findings accord with papers [30], [45], [65], and [71] who reported 1-D CNN can yield better results than the common DL techniques in certain conditions. In addition, the results amplify the importance of dividing reference datasets into three (training, validation, and test datasets) so that the performance of a model is strictly evaluated on unseen data.

In other words, reusing the validation datasets that have been used in model parameter tuning for testing the generalizability could lead to bias or incorrect conclusions due to information leakage during parameter tuning operations [30], [64].

Moreover, achieving the best and the second-best results using RF and 1-D CNN models respectively indicates the superiority of pixel-based classification that considers only spectral values over the patch-based DL techniques that consider the contextual/spatial. This further implies that spatial information has little/no significance in coarse-resolution image classification. In other words, spectral information is more important than spatial information in coarse-resolution image classification as stated by the authors [30], [66], [72]. To put it simply, we cannot use what we barely have. CNN cannot reveal information that is not contained in the image [23].

In low-resolution images, spatial details such as texture, shape, and patterns specific to different classes are often not easily identifiable. This is because the sensors homogenize all information below their spatial resolution into the dominant classes leading to biases and misclassifications, particularly in areas where land cover varies within small distances. Essentially, the spatial details are significantly compromised or lost in low-resolution images [72], forcing CNN and segmentation models to rely heavily on spectral information rather than distinct spatial features. This lack of clear spatial information may introduce biases, negatively impacting model performance and necessitating additional training data. This could explain why pixel-based classification methods have outperformed

TABLE IV

Data type	model	chip size	Accuracy , support	Bare-Sparse vegetation	Built up	Cropland	Forest	Herbaceous wetland	Herbaceous vegetation	Shrub	Water body	
Valid ation	1D CNN	-	support	4384	252	3472	2127	291	2343	4372	2816	
			precision	0.95	0.77	0.87	0.90	0.85	0.74	0.85	1.00	
			recall	0.95	0.67	0.90	0.88	0.85	0.85	0.78	1.00	
			f1-score	0.95	0.71	0.89	0.89	0.85	0.79	0.81	1.00	
			OA	0.89								
			kapa(k)	0.86								
Valid ation	2D CNN	2x2	support	1212	95	1002	625	98	675	1264	775	
			precision	0.91	0.59	0.90	0.91	0.51	0.65	0.76	0.99	
			recall	0.99	0.52	0.75	0.83	0.86	0.73	0.74	0.99	
			f1-score	0.95	0.55	0.82	0.87	0.64	0.69	0.75	0.99	
			OA	0.84								
			kapa(k)	0.81								
Valid ation	2D CNN	3x3	support	595	57	519	329	52	337	640	381	
			precision	0.90	0.81	0.88	0.79	0.43	0.56	0.84	0.99	
			recall	0.98	0.51	0.77	0.84	0.73	0.70	0.70	0.98	
			f1-score	0.94	0.62	0.82	0.82	0.54	0.62	0.76	0.99	
			OA	0.82								
			kapa(k)	0.79								
Valid ation	2D CNN	5x5	support	257	37	240	154	31	156	293	157	
			precision	0.80	0.57	0.85	0.75	0.37	0.58	0.73	0.91	
			recall	0.96	0.35	0.71	0.82	0.48	0.62	0.62	0.99	
			f1-score	0.87	0.43	0.78	0.79	0.42	0.60	0.67	0.95	
			OA	0.76								
			kapa(k)	0.71								
Valid ation	3D CNN	2x2	support	1212	95	1002	625	98	675	1264	775	
			precision	0.94	0.77	0.88	0.85	0.76	0.67	0.80	1.00	
			recall	0.97	0.77	0.79	0.84	0.77	0.82	0.74	1.00	
			f1-score	0.95	0.77	0.83	0.85	0.76	0.73	0.77	1.00	
			OA	0.85								
			kapa(k)	0.82								
Valid ation	U-net	3x3	support	4384	252	3472	2125	288	2343	4372	2816	
			precision	0.94	0.54	0.78	0.91	0.61	0.66	0.81	0.98	
			recall	0.95	0.20	0.90	0.82	0.80	0.75	0.69	0.99	
			f1-score	0.95	0.29	0.83	0.86	0.69	0.70	0.74	0.98	
			OA	0.84								
			kapa(k)	0.81								
Valid ation	U-net	5x5	support	4384	252	3472	2123	288	2343	4372	2816	
			precision	0.93	0.68	0.84	0.70	0.72	0.63	0.71	0.98	
			recall	0.99	0.25	0.74	0.87	0.82	0.73	0.61	1.00	
			f1-score	0.96	0.37	0.79	0.77	0.76	0.67	0.65	0.99	
			OA	0.81								
			kapa(k)	0.77								
Valid ation	RF	-	support	4384	252	3472	2127	291	2343	4372	2816	
			precision	0.95	0.48	0.95	0.73	0.80	0.69	0.84	1.00	
			recall	0.94	0.63	0.78	0.95	0.79	0.89	0.71	1.00	
			f1-score	0.94	0.55	0.86	0.82	0.80	0.77	0.77	1.00	
			OA	0.86								
			kapa(k)	0.83								

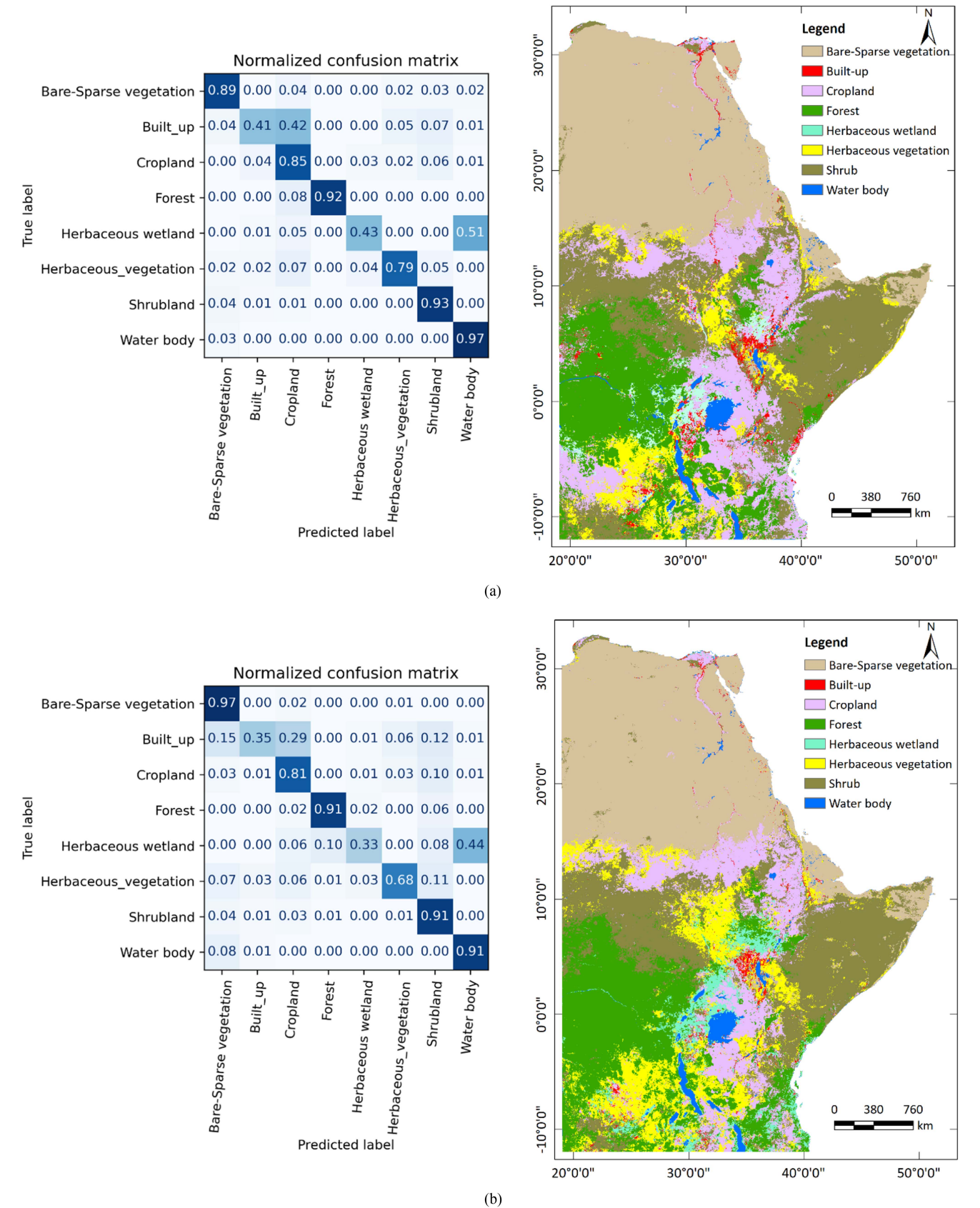
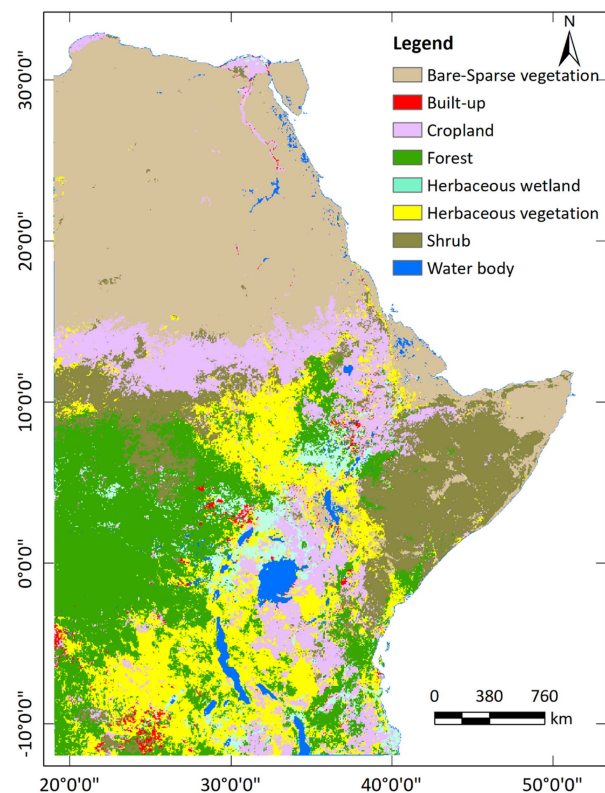
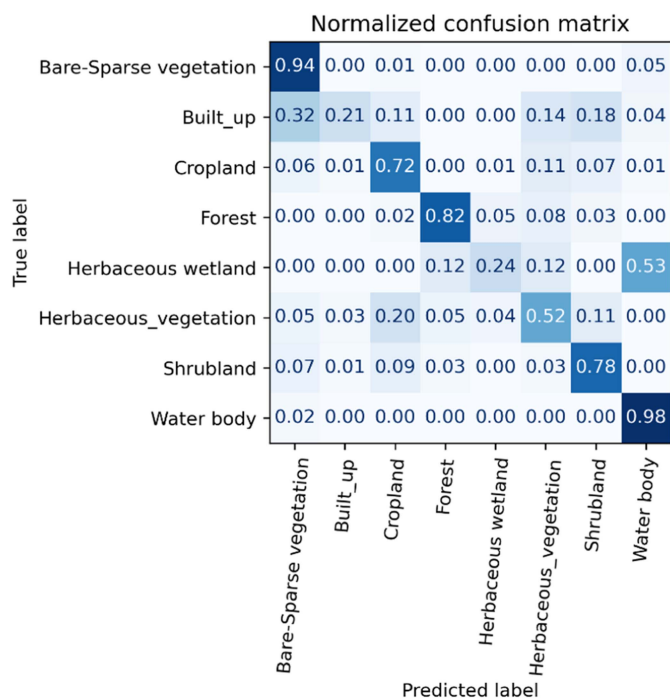
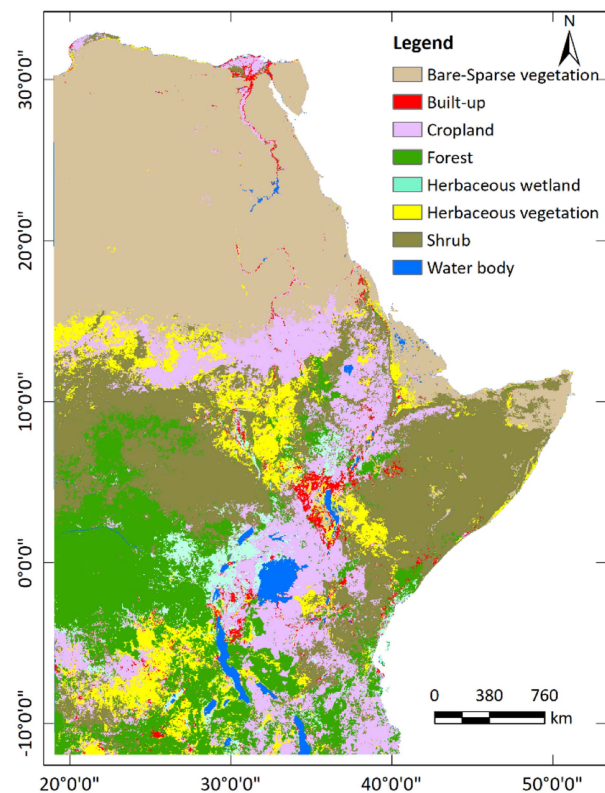
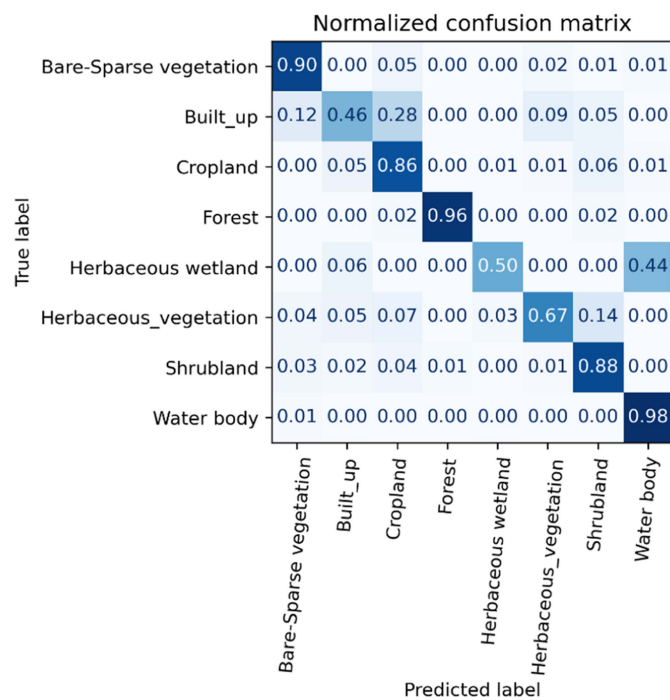


Fig. 9. Confusion matrix generated using test datasets for the various CNNs, U-net, and ML models and the corresponding final land cover products. (a) 1-D CNN. (b) 2-D CNN using 2 × 2 input patch.

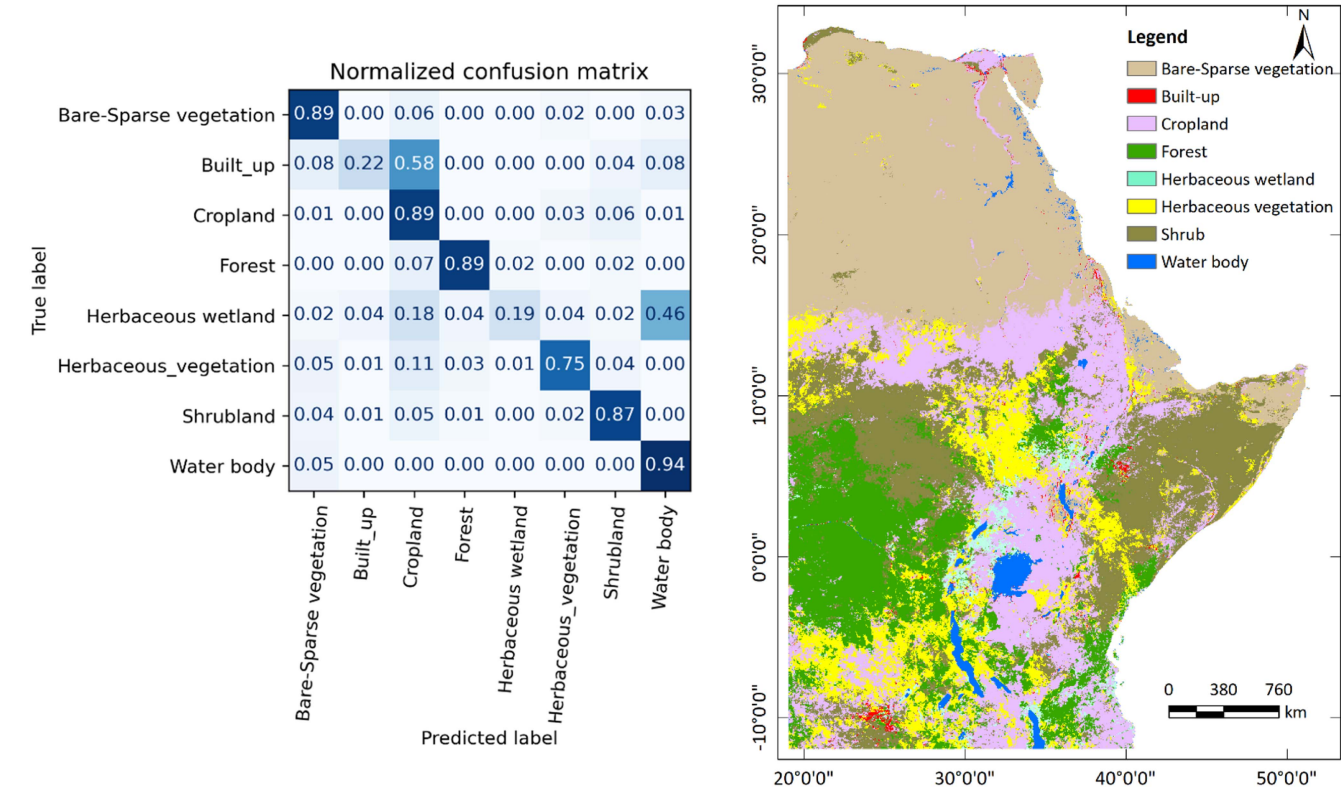


(c)

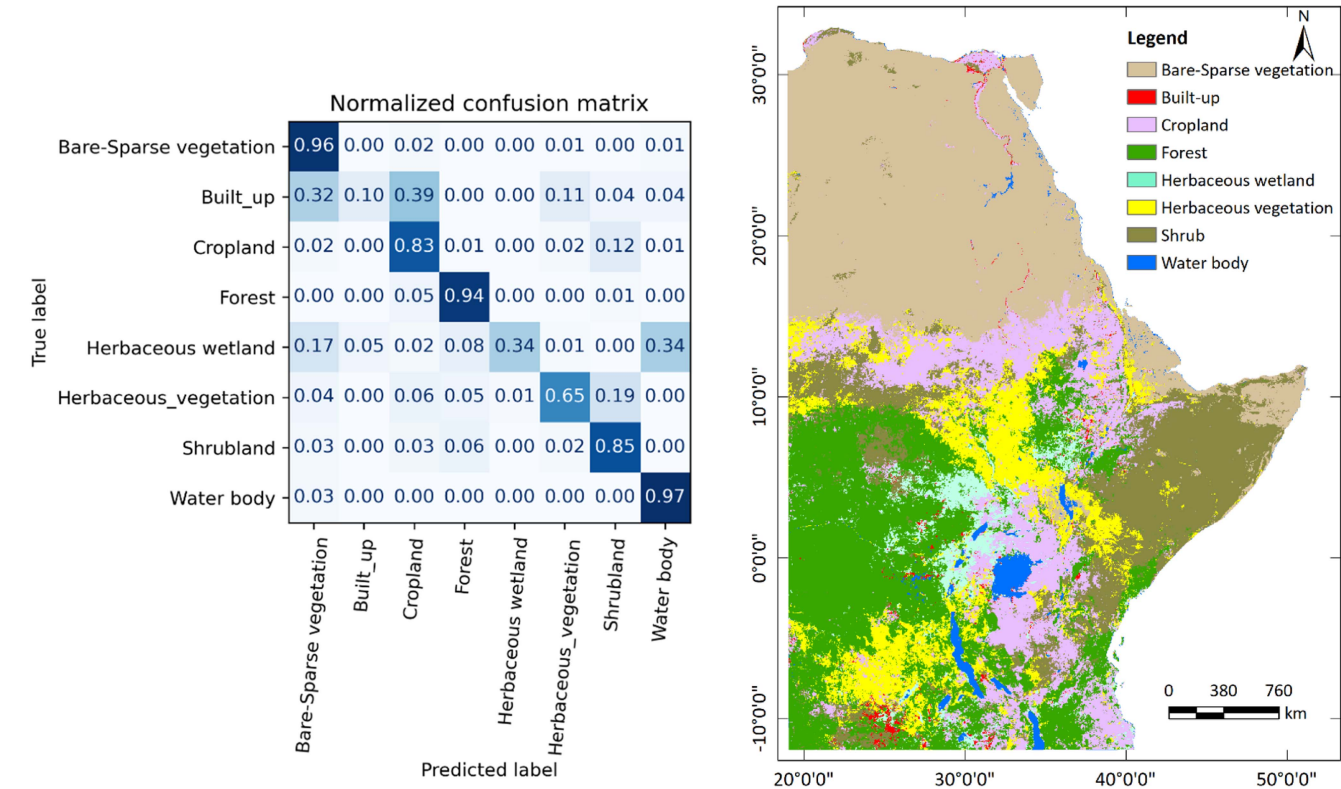


(d)

Fig. 9. (Continued.) Confusion matrix generated using test datasets for the various CNNs, U-net, and ML models and the corresponding final land cover products. (c) 2-D CNN using 5×5 patch. (d) 3-D CNN using 2×2 input size.



(e)



(f)

Fig. 9. (Continued.) Confusion matrix generated using test datasets for the various CNNs, U-net, and ML models and the corresponding final land cover products. (e) U-net using 3 × 3 input chips. (f) U-net using 5 × 5 input chips.

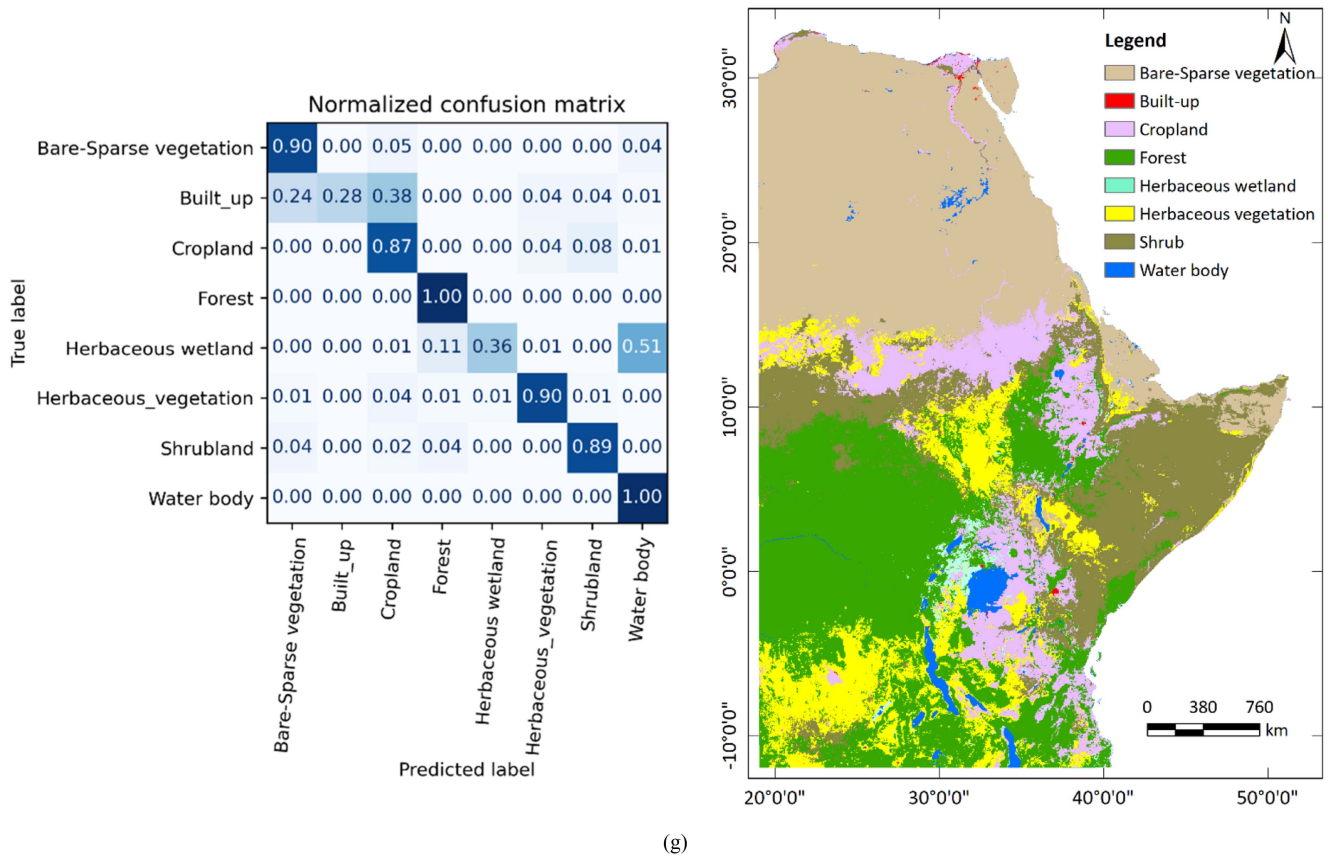


Fig. 9. (Continued.) Confusion matrix generated using test datasets for the various CNNs, U-net, and ML models and the corresponding final land cover products. (g) RF.

patch-based CNN models (2-D and 3-D) and segmentation models.

The other reason could be the quantity of the reference data. Patch-based CNN typically requires more reference datasets than pixel-based techniques [6], [45]. In this regard, the need for more training data might be more significant in coarse resolution than high resolution due to the less distinct spatial information in coarse images.

On the other hand, the possible explanation for why segmentation could not achieve the best result being a pixel-wise classification technique is that its initial classification stage involves segmentation of the classes into categories using the convolving kernels, and CNN layers operation. And, that could also be the reason why the CNN and U-net models achieved similar accuracy when smaller chip size samples were used (see Tables IV and V), despite differences in performance across individual classes and the final land cover maps (see Fig. 9 column 2).

B. Impact of Input Patch Size

For the patch-based classification and U-net models, the impacts of chip size on accuracy were evaluated by considering different chip sizes 2×2 , 3×3 , and 5×5 . The results demonstrated that chip size is inversely related to accuracy.

As shown in Tables IV and V, 2-D CNN achieved its maximum accuracy ($OA = 0.85$ and $k = 0.82$) when the smallest chip size was used, and 1%–2% lesser accuracy when 3×3 chips were used; however, it yielded the least accuracy, which is 9% lesser, when the chip size increased to 5×5 . Similarly, the accuracy of the U-net model decreased by 3% on validation datasets and 1% on tests when the size of patches increased from 3×3 to 5×5 . This inverse relation could be caused by one or the combination of the following two factors. The first factor is, increasing the patch size in a situation where a low-resolution image (1 km) was used for extensive heterogeneous area mapping, as in our case, increases intraclass and interclass heterogeneity, i.e., variation in spectral and spatial information of the land cover types that leads to confusion between classes and low accuracy. The other reason is, which is particularly related to the CNN models, chipping the images into larger sizes means reducing the total amount of reference data (see Table II), as a result, the models fail to learn complete information in the data and yield low accuracy.

Previously, some researchers have demonstrated the effects of patch size. Consistent with our findings, Paoletti et al. [65], and Meng et al. [73] reported that larger patch sizes lead to lower accuracy.

In addition to negatively impacting the performance of the models, utilizing larger patches in patch-based CNN models for coarse-resolution image classification can lead to

extraneous outcomes, including missing minor classes whose spatial dimension/areal extent is less than the patch size, and/or losing their real shape either partially or completely. For instance, as shown in Fig. 9(c), processing 5×5 chips using 2-D CNN resulted in missing the Congo River, i.e., the Congo River in the central Congo was left undetected as it was homogenized to the majority neighboring class, to the forest, during the classification of the entire image. Similarly, the trace of the Blue Nile River has been partially missed, particularly the southern part of the river. However, with similar patch sizes, the problem is less pronounced in U-net products [see Fig. 9(f)]. Nonetheless, the rivers and/or their traces were detected when the smallest chips were processed [see Fig. 9(b), (d), and (e)]. Furthermore, they were mapped more clearly when pixel-wise classification (RF, 1-D CNN) was employed [see Fig. 9(a), (e), and (g)], indicating the benefit of employing pixel-based classification in coarse-resolution image classification besides achieving high accuracy.

VI. CONCLUSION

In this work, we assessed the effectiveness and applicability of the most commonly used DL models, such as 1-D CNN, 2-D CNN, 3-D CNN, and U-net semantic segmentation models for coarse resolution image classification to generate regional land cover maps of a part of Africa. In addition, we compared these models with one of the most robust conventional ML classifiers, the RF, to evaluate their performance and feasibility.

We processed 9 months, coarse-resolution (1km), 10-day composite FY-3C imageries comprising 11 bands. Labeled datasets were collected in shapefiles/ROI format and split into three independent datasets: train, validation, and test datasets. Then, the datasets were rasterized and preprocessed to meet each model's reference dataset format/shape requirement. Python programming language is the main tool employed to manipulate the datasets.

Our study revealed that DL techniques can be effectively used for coarse-resolution image classification. However, selecting the right model and/or chip size is a crucial precondition to obtaining the best result, as models' performance is highly affected by the type of models employed and/or the size of the patches processed, particularly for patch-based models. Among the DL techniques, 1-D CNN the pixel-based approach outperformed all the patch-based models (2-D, 3-D,) and U-net segmentation techniques but was outperformed by RF; hence, 1-D CNN is a more feasible DL technique compared to patch-based and U-net DL models, yet less effective than RF in coarse-resolution image classification.

Regarding the performance of patch-based (2-D CNN and 3-D CNN) and U-net segmentation models, the models demonstrated comparable high accuracy when small patch-size reference datasets are utilized; however, the performance of patch-based models is adversely impacted as the tile/chip size increases. Additionally, employing larger patches in coarse-resolution image classification leads to missing some smaller classes, partially or entirely, resulting in misclassification.

Furthermore, achieving the highest and second-highest accuracy using RF and 1-D CNN models, respectively, highlights the

superiority of pixel-based classification that focuses solely on spectral values over patch-based DL techniques that incorporate both contextual/spatial and spectral information. This further suggests the limited significance of spatial information in coarse-resolution image classification.

However, although the DL models, CNN, and U-net can achieve high accuracy, especially 1-D CNN, they are less feasible and suitable than RF for coarse resolution image classification of big datasets, particularly in extensive area land cover mapping. This is not only because the RF models outperformed them, but also due to other critical reasons such as the complexity of the DL models and procedures involved. These include intricate pre and postprocessing of datasets, having numerous parameters and hyperparameters requiring tuning, which is daunting, and being prone to overfitting and computationally and memory-expensive. While RF can achieve the best results using default parameters as given in the sklearn python library besides its simplicity.

In this study, we split the reference datasets into three independent sets, to train, validate, and test the models. The method is the best approach to strictly evaluate the models [64], especially if repetitive model training is conducted to finetune various hyperparameters and obtain/select the best models. However, this type of evaluation protocol is recommended if sufficient datasets are available; unless it reduces the amount of samples and impacts the accuracy. That is one of the reasons why all the models yielded the least accuracy (precision, recall, and F1) on the minor classes (built-up and/or herbaceous wetland), classes with a small number of samples.

To overcome challenges associated with limited reference datasets, [64] suggested using cross-validation techniques instead of dividing the reference datasets into three. However, in our case, the technique was ineffective for two reasons: the nature of our labeled samples and class imbalance.

First, since our labeled datasets were collected as shapefile/ROI consisting of different-size polygons, it is highly likely that a single polygon in a rasterized shapefile is chipped into two or more chips/parts. Therefore, when the chips are split randomly into training and validation samples, there is a high probability that chips from the same polygon are divided into training and validation sets, i.e., there is a correlation between training and validation samples. This leads to validating the model on some of the datasets it was trained on, resulting in biased results/incorrect validation.

The other reason is class imbalances. Utilizing unequal amounts of reference datasets for each class affects the probability and amount of each class that goes into each fold while splitting the data randomly into a number of folds. This also leads to biased validation results where the dominant/the most frequent classes are predicted with high accuracy while the minor classes have low accuracy. Therefore, generating sufficient and balanced reference datasets across all classes is crucial to effectively split the datasets into three independent sets, or collecting balanced labeled ROI/shapefiles with equal size polygons is recommended to implement cross-validation techniques in case of sparse datasets.

ACKNOWLEDGMENT

The authors are grateful to the China Meteorological Administration, the National Satellite Meteorological Center for providing the FY-3C data free of charge.

REFERENCES

- [1] M. A. Friedl et al., "MODIS collection 5 global land cover: Algorithm refinements and characterization of new datasets," *Remote Sens. Environ.*, vol. 114, no. 1, pp. 168–182, 2010.
- [2] F. Sedano, P. Gong, and M. Ferrao, "Land cover assessment with MODIS imagery in southern African Miombo ecosystems," *Remote Sens. Environ.*, vol. 98, no. 4, pp. 429–441, 2005.
- [3] O. Arino, P. Bicheron, F. Achard, J. Latham, R. Witt, and J.-L. Weber, "The most detailed portrait of Earth," *Eur. Space Agency*, vol. 136, pp. 25–31, 2008.
- [4] P. Mayaux, E. Bartholomé, S. Fritz, and A. Belward, "A new land-cover map of Africa for the year 2000," *J. Biogeogr.*, vol. 31, no. 6, pp. 861–877, 2004.
- [5] G. Camps-Valls and L. Bruzzone, *Kernel Methods for Remote Sensing Data Analysis*, 1st ed. Hoboken, NJ, USA: Wiley, 2009, p. 434.
- [6] J. Song, S. Gao, Y. Zhu, and C. Ma, "A survey of remote sensing image classification based on CNNs," *Big Earth Data*, vol. 3, no. 3, pp. 232–254, 2019.
- [7] A. E. Maxwell, T. A. Warner, and F. Fang, "Implementation of machine-learning classification in remote sensing: An applied review," *Int. J. Remote Sens.*, vol. 39, no. 9, pp. 2784–2817, 2018.
- [8] J. B. Campbell and R. H. Wynne, *Introduction to Remote Sensing*, 5th ed. New York, NY, USA: Guilford Press, 2011.
- [9] M. A. Friedl and C. E. Brodley, "Decision tree classification of land cover from remotely sensed data," *Remote Sens. Environ.*, vol. 61, no. 3, pp. 399–409, 1997.
- [10] M. C. Hansen and B. Reed, "A comparison of the IGBP DISCover and University of Maryland 1 km global land cover products," *Int. J. Remote Sens.*, vol. 21, nos. 6/7, pp. 1365–1373, 2000.
- [11] C. Huang, L. Davis, and J. Townshend, "An assessment of support vector machines for land cover classification," *Int. J. Remote Sens.*, vol. 23, no. 4, pp. 725–749, 2002.
- [12] M. Pal, "Random forest classifier for remote sensing classification," *Int. J. Remote Sens.*, vol. 26, no. 1, pp. 217–222, 2005.
- [13] M. Pal and P. M. Mather, "Support vector machines for classification in remote sensing," *Int. J. Remote Sens.*, vol. 26, no. 5, pp. 1007–1011, 2005.
- [14] B. Ghimire, J. Rogan, V. R. Galiano, P. Panday, and N. Neeti, "An evaluation of bagging, boosting, and random forests for land-cover classification in cape cod, Massachusetts, USA," *GIScience Remote Sens.*, vol. 49, no. 5, pp. 623–643, 2012.
- [15] J. R. Otukei and T. Blaschke, "Land cover change assessment using decision trees, support vector machines and maximum likelihood classification algorithms," *Int. J. Appl. Earth Observation Geoinf.*, vol. 12, pp. S27–S31, 2010.
- [16] P. Gong et al., "Finer resolution observation and monitoring of global land cover: First mapping results with landsat TM and ETM+ data," *Int. J. Remote Sens.*, vol. 34, no. 7, pp. 2607–2654, 2013.
- [17] Y. Ren et al., "Full convolutional neural network based on multi-scale feature fusion for the class imbalance remote sensing image classification," *Remote Sens.*, vol. 12, no. 21, 2020, Art. no. 3547.
- [18] J. V. Solórzano, J. F. Mas, Y. Gao, and J. A. Gallardo-Cruz, "Land use land cover classification with u-net: Advantages of combining sentinel-1 and sentinel-2 imagery," *Remote Sens.*, vol. 13, no. 18, 2021, Art. no. 3600.
- [19] S. S. Heydari and G. Mountrakis, "Meta-analysis of deep neural networks in remote sensing: A comparative study of mono-temporal classification to support vector machines," *ISPRS J. Photogrammetry Remote Sens.*, vol. 152, pp. 192–210, Jun. 2019.
- [20] A. Sharma, X. Liu, X. Yang, and D. Shi, "A patch-based convolutional neural network for remote sensing image classification," *Neural Netw.*, vol. 95, pp. 19–28, Nov. 2017.
- [21] E. Saralioglu and O. Gungor, "Semantic segmentation of land cover from high resolution multispectral satellite images by spectral-spatial convolutional neural network," *Geocarto Int.*, vol. 37, no. 2, pp. 657–677, Jan. 2022.
- [22] L. Zhang, L. Zhang, and B. Du, "Deep learning for remote sensing data: A technical tutorial on the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 22–40, Jun. 2016.
- [23] T. Kattenborn, J. Leitloff, F. Schiefer, and S. Hinz, "Review on convolutional neural networks (CNN) in vegetation remote sensing," *ISPRS J. Photogrammetry Remote Sens.*, vol. 173, pp. 24–49, Mar. 2021.
- [24] L. Alzubaidi et al., "Review of deep learning: Concepts, CNN architectures, challenges, applications, future directions," *J. Big Data*, vol. 8, no. 1, Mar. 2021, Art. no. 53.
- [25] A. Stoian, V. Poulain, J. Inglada, V. Poughon, and D. Derksen, "Land cover maps production with high resolution satellite image time series and convolutional neural networks: Adaptations and limits for operational systems," *Remote Sens.*, vol. 11, no. 17, 2019, Art. no. 1986.
- [26] W. Boonpook et al., "Deep learning semantic segmentation for land use and land cover types using landsat 8 imagery," *ISPRS Int. J. Geo-Inf.*, vol. 12, no. 1, 2023, Art. no. 14.
- [27] O. Sen and H. Y. Keles, "On the evaluation of CNN models in remote-sensing scene classification domain," *PFG, J. Photogrammetry, Remote Sens. Geoinf. Sci.*, vol. 88, no. 6, pp. 477–492, Dec. 2020.
- [28] P. Zhang, Y. Ke, Z. Zhang, M. Wang, P. Li, and S. Zhang, "Urban land use and land cover classification using novel deep learning models based on high spatial resolution satellite imagery," *Sensors*, vol. 18, no. 11, 2018, Art. no. 3717.
- [29] F. Zhao et al., "Monthly mapping of forest harvesting using dense time series sentinel-1 SAR imagery and deep learning," *Remote Sens. Environ.*, vol. 269, Feb. 2022, Art. no. 112822.
- [30] J. Liu, T. Wang, A. Skidmore, Y. Sun, P. Jia, and K. Zhang, "Integrated 1D, 2D, and 3D CNNs enable robust and efficient land cover classification from hyperspectral imagery," *Remote Sens.*, vol. 15, no. 19, 2023, Art. no. 4797.
- [31] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017.
- [32] Y. Song et al., "Land cover classification for satellite images through 1D CNN," in *Proc. Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens.*, 2019, pp. 1–5.
- [33] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chansussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.
- [34] J. Yao, D. Hong, C. Li, and J. Chansussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.
- [35] C. F. Brown et al., "Dynamic world, near real-time global 10 m land use land cover mapping," *Sci. Data*, vol. 9, no. 1, Jun. 2022, Art. no. 251.
- [36] K. Karra, C. Kontgis, Z. Statman-Weil, J. C. Mazzariello, M. Mathis, and S. P. Brumby, "Global land use /land cover with sentinel 2 and deep learning," in *Proc. 2021 IEEE Int. Geosci. Remote Sens. Symp.*, 2021, pp. 4704–4707.
- [37] M. A. Wulder, N. C. Coops, D. P. Roy, J. C. White, and T. Hermosilla, "Land cover 2.0," *Int. J. Remote Sens.*, vol. 39, no. 12, pp. 4254–4284, 2018.
- [38] W. Ballard, *Hands-on Deep Learning for Images With TensorFlow: Build Intelligent Computer Vision Applications Using TensorFlow and Keras*. Birmingham, U.K.: Packt Publishing, 2018.
- [39] S. Minaee, Y. Boykov, F. Porikli, A. Plaza, N. Kehtarnavaz, and D. Terzopoulos, "Image segmentation using deep learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3523–3542, Jul. 2022.
- [40] S. Patel, "Deep learning models for image segmentation," in *Proc. 8th Int. Conf. Comput. Sustain. Glob. Develop.*, 2021, pp. 149–154.
- [41] O. Köpüklü, N. Köse, A. Gunduz, and G. Rigoll, "Resource efficient 3D convolutional neural networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop*, Oct. 2019, pp. 1910–1919.
- [42] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.
- [43] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Convolutional neural networks for large-scale remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 645–657, Feb. 2017.
- [44] N. Singh and H. Sabrol, "Convolutional neural networks-an extensive arena of deep learning. A comprehensive study," *Arch. Comput. Methods Eng.*, vol. 28, no. 7, pp. 4755–4780, Dec. 2021.
- [45] S. Kiranyaz, O. Avci, O. Abdeljaber, T. Ince, M. Gabbouj, and D. J. Inman, "1D convolutional neural networks and applications: A survey," *Mech. Syst. Signal Process.*, vol. 151, Apr. 2021, Art. no. 107398.
- [46] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.
- [47] Q. Yuan et al., "Deep learning in environmental remote sensing: Achievements and challenges," *Remote Sens. Environ.*, vol. 241, May 2020, Art. no. 111716.

- [48] X. Zhang, Y. N. Zhou, and J. Luo, "Deep learning for processing and analysis of remote sensing Big Data: A technical review," *Big Earth Data*, vol. 6, no. 4, pp. 527–560, Oct. 2022.
- [49] N. Ketkar, *Deep Learning With Python: A Hands-on Introduction*. Berkeley, CA, USA: Apress, 2017.
- [50] G. Litjens et al., "A survey on deep learning in medical image analysis," *Med. Image Anal.*, vol. 42, pp. 60–88, Dec. 2017.
- [51] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [52] A. Clark, S. Phinn, and P. Scarth, "Optimised U-net for land use–land cover classification using aerial photography," *PFG, J. Photogrammetry, Remote Sens. Geoinf. Sci.*, vol. 91, no. 2, pp. 125–147, Apr. 2023.
- [53] J. Ball, D. Anderson, and C. S. Chan, "A comprehensive survey of deep learning in remote sensing: Theories, tools and challenges for the community," *J. Appl. Remote Sens.*, vol. 11, pp. 13–16, Aug. 2017.
- [54] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 3431–3440.
- [55] L. Ma, Y. Liu, X. Zhang, Y. Ye, G. Yin, and B. A. Johnson, "Deep learning in remote sensing applications: A meta-analysis and review," *ISPRS J. Photogrammetry Remote Sens.*, vol. 152, pp. 166–177, Jun. 2019.
- [56] Z. Wu et al., "Deep learning enables satellite-based monitoring of large populations of terrestrial mammals across heterogeneous landscape," *Nature Commun.*, vol. 14, no. 1, May 2023, Art. no. 3072.
- [57] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.
- [58] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.
- [59] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Medical Image Comput. Computer-Assisted Intervention*, Oct. 2015, pp. 234–241.
- [60] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
- [61] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogrammetry Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.
- [62] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 5686–5696.
- [63] T. Adugna, W. Xu, and J. Fan, "Effect of using different amounts of multi-temporal data on the accuracy: A case of land cover mapping of parts of africa using FengYun-3C data," *Remote Sens.*, vol. 13, no. 21, 2021, Art. no. 4461.
- [64] F. Chollet, *Deep Learning With Python*. New York, NY, USA: Simon and Schuster, 2021.
- [65] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogrammetry Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.
- [66] T. Kattenborn, J. Eichel, and F. E. Fassnacht, "Convolutional neural networks enable efficient, accurate and fine-grained segmentation of plant species and communities from high-resolution UAV imagery," *Sci. Rep.*, vol. 9, no. 1, Nov. 2019, Art. no. 17656.
- [67] M. Belgiu and L. Drăguț, "Random forest in remote sensing: A review of applications and future directions," *ISPRS J. Photogrammetry Remote Sens.*, vol. 114, pp. 24–31, 2016.
- [68] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, Oct. 2001, doi: [10.1023/A:1010933404324](https://doi.org/10.1023/A:1010933404324).
- [69] T. Adugna, W. Xu, and J. Fan, "Comparison of random forest and support vector machine classifiers for regional land cover mapping using coarse resolution FY-3C images," *Remote Sens.*, vol. 14, no. 3, 2022, Art. no. 574.
- [70] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.
- [71] D. B. Sencaki et al., "Land cover multiclass classification of wonosobo, Indonesia with time series-based one-dimensional deep learning model," *Remote Sens. Appl., Soc. Environ.*, vol. 32, 2023, Art. no. 101040.
- [72] D. Lu and Q. Weng, "A survey of image classification methods and techniques for improving classification performance," *Int. J. Remote Sens.*, vol. 28, no. 5, pp. 823–870, 2007.
- [73] Y. Meng, Z. Ma, Z. Ji, R. Gao, and Z. Su, "Fine hyperspectral classification of rice varieties based on attention module 3D-2DCNN," *Comput. Electron. Agriculture*, vol. 203, Dec. 2022, Art. no. 107474.



Tesfaye Adugna received the B.Sc. degree in earth sciences (geology) with a major in geology and a minor in chemistry, and the M.Sc. degree in earth sciences (structural geology) from Addis Ababa University, Addis Ababa, Ethiopia, in 2005 and 2014, respectively, and the Ph.D. degree in information and communication engineering, focusing on satellite image processing and application, from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in June 2022.

With over ten years of experience, he has worked as a Researcher and Lecturer with Addis Ababa Science and Technology University and as a Geologist/Exploration Geologist in various regions of Ethiopia. He has been working as a Scientific Researcher with the Yangtze Delta Region Institute (Huzhou), UESTC, since August 2022. His research interests include remote sensing/satellite image processing, machine learning, deep learning, semantic segmentation, land cover mapping, application of remote sensing in geosciences, mineral exploration, structural geology, environmental impact analysis, and GIS.



Wenbo Xu received the B.S. degree in biomedical engineering from Chongqing University, Chongqing, China, in 1994, the M.S. degree in communication and information systems from the Chongqing University of Posts and Telecommunications, Chongqing, in 1998, and the Ph.D. degree in cartography and geographic information systems from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, China, in 2004.

From 2004 to 2012, he was a Lecturer and Associate Professor with the School of Automation Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China. From 2013 to 2014, he was a Visiting Scholar with the Department of Geography, University of Iowa, Iowa City, USA. Since 2014, he has been a Professor with the School of Resources and Environment, UESTC. He has authored 1 book, more than 50 articles, and more than 10 inventions. His research interests include environment remote sensing, image processing and application, and geographic information systems. He holds three patents.

Dr. Xu was a recipient of the first prize of the Chinese Science and Technology Progress Award of Surveying and Mapping in 2019, the first prize of the Science and Technology Progress Award of Surveying and Mapping Geographic Information of Sichuan Province in 2017, and the second prize of Science and Technology Progress Award of Sichuan Province in 2020.



Jinlong Fan received the B.Sc. degree in soil and plant nutrition from Shanxi Agricultural University, Jinzhong, China, in 1996, the M.Sc. degree in agricultural meteorology from China Agricultural University, Beijing, China, in 2000, and the Ph.D. degree in agricultural remote sensing from the Institute of Remote Sensing Applications, Chinese Academy of Sciences, Beijing, in 2003, respectively.

Dr. Fan is currently a professor with the faculty of Geographical Sciences, Beijing Normal University. He had ever worked as a research professor for 21 years with National Satellite Meteorological Center, China Meteorological Administration before he joined the Beijing Normal University. His research interests include the processing of satellite data, quick responses to disaster monitoring and agricultural monitoring with remote sensing.



Haitao Jia received the Ph.D. degree in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2012.

He is currently a Professor with the School of Resources and Environment, UESTC. He is also working with the Yangtze Delta Region Institute (Huzhou), UESTC. He has authored or coauthored more than 40 research papers. His research interests include intelligent information processing, image processing, machine learning, and artificial intelligence, and Big

Data processing.

Dr. Jia was the recipient of the third prize of the National Defense Science and Technology Progress Award in 2009, the second prize of the Sichuan Science and Technology Progress Award in 2013, the second prize of the Science and Technology Award of the Chinese Society of Surveying and Mapping in 2020, the second prize of computer application innovation product China computer federation.



Xin Luo was born in Chengdu, Sichuan, China, in 1977. She received the B.E. degree in electrical technology, the M.E. degree in power electronics and electric drive, and the Ph.D. degree in pattern recognition and intelligent systems from Northwestern Polytechnical University, Xi'an, China, in 1999, 2004, and 2008, respectively.

From 2008 to 2012, she was a Lecturer with the School of Communication and Information Engineering, the University of Electronic Science and Technology of China (UESTC), Chengdu, China. Since

2012, she has been an Associate Professor with the School of Resources and Environment, UESTC. From 2014 to 2015, she was a Visiting Scholar with the Department of Mathematics, University of California, Santa Barbara, CA, USA. She is currently working as a Scientific Researcher with the Yangtze Delta Region Institute, UESTC, Huzhou, China. She has authored more than one book and more than ten inventions. Her research interests include remote sensing image processing and application, machine learning, pattern recognition, computer vision, and 3-D modeling.

Dr. Luo was the first recipient of the IEEE International Conference on Knowledge Innovation and Invention Best Conference Paper Award in 2018, the second prize of Science and Technology Award of Surveying and Mapping Geographic Information of China in 2020, and the third prize of Science and Technology Progress Award of Sichuan Province in 2021, respectively. She is an expert in the Sichuan Branch of the Chinese National Remote Sensing Center and a member of the Chinese Society for Geodesy Photogrammetry and Cartography.